

2

Towards the Modern Atomic Model

In this chapter, we will learn about

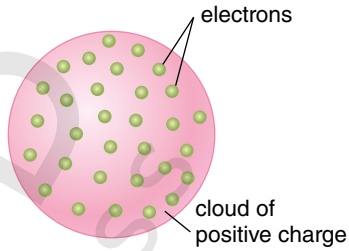
- Rutherford's atomic model and its limitations
- the line spectra and evidence of energy levels of atoms
- Bohr's model of the hydrogen atom and how it explains the line spectra
- the wave-particle duality of light and electrons

2.1 Rutherford's model and scattering experiment



J. J. Thomson.

In an atom, the tiny electrons are embedded in a cloud of positive charge, like a plum pudding.





Hans Geiger.



Ernest Marsden.

We found that some α particles bounce back when they hit a gold foil.

What! If the positive charge spreads out over the atom's entire volume, it will not be able to bounce back the α particles. It is like firing a 15-inch shell at a piece of tissue paper and it came back and hit you!



Ernest Rutherford.

The findings of Geiger and Marsden inspired Rutherford to modify Thomson's plum-pudding atomic model. Let us learn more about this.

1 Geiger-Marsden experiment

In the Geiger-Marsden experiment, a thin gold foil was bombarded with α particles in a vacuum (Fig 2.1a). Scattering of α particles occurred.

A gold foil was used because it can be made very thin so that each α particle would most likely be scattered by one gold nucleus only.

Scattering is a process in which particles or waves are deflected from the original paths by a force field.

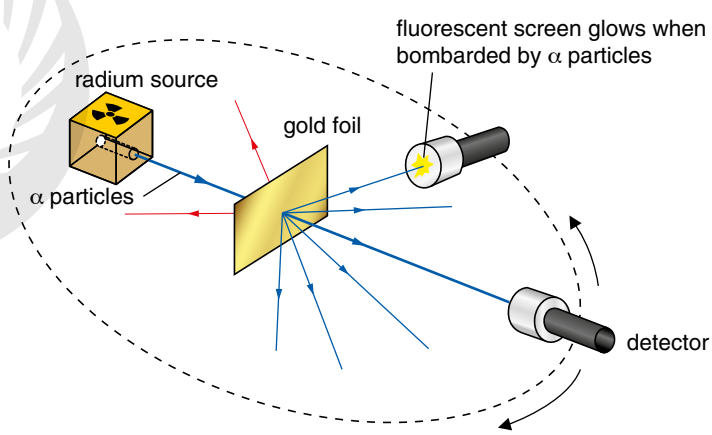


Fig 2.1a *The Geiger-Marsden experiment.*

Know more

Ernest Rutherford (1871–1937)



In addition to his work on the atomic model, Rutherford had done a lot of important work on the disintegration of elements and the chemistry of radioactive substances. He received the Nobel Prize in Chemistry in 1908. He is known as the 'father of nuclear physics'.

OUP eResources

J. J. Thomson 湯姆生　Hans Geiger 蓋革　Ernest Marsden 馬斯登　Ernest Rutherford 盧瑟福

The following results were obtained:

1 Nearly all the α particles passed straight through the foil.

2 A very small number of α particles were deflected and a few (about 1 in 8000) even bounced back.

The second result could not be explained by Thomson's model. In this model, the positive charge distributed all over an atom like a cloud and could not provide sufficient electric force to bounce back an α particle.

DSE goal

Describe Rutherford's construction of an atomic model consisting of a nucleus and electrons.

DSE exam

15(2)Q2.1, 16(2)Q2.1, 17(2)Q2.1, 18(2)Q2.1, 19(2)Q.2(a), 20(2)Q2.1, 21(2)Q2.1, 23(2)Q2.1, etc.

2 Rutherford's atomic model

In order to explain the experimental results, Rutherford proposed another atomic model (Fig 2.1b) which has the following features:

All the positive charge and most of the mass of the atom are concentrated in a tiny core called the nucleus.

The electric force between an electron and the nucleus keeps the electron orbiting the nucleus along a circular path.

The rest of the atom is largely empty space.

Rutherford estimated that if an atom were the size of a football pitch, its nucleus would be no bigger than a pea!



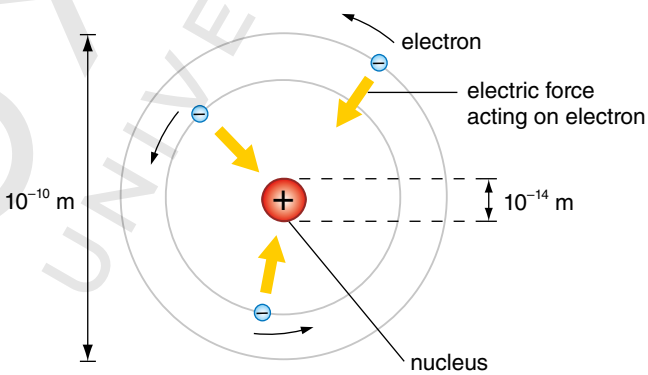


Fig 2.1b *Rutherford's atomic model (not drawn to scale).*

Ss may have a common misconception that scientific models are what really exist in nature. Some may even think that scientists have seen electrons orbiting around the nucleus of an atom and hence drawn what they have seen. Ts may emphasize that scientific models are scientists' creation in an attempt to explain phenomena. Model building is an important activity through which science progresses.

Nature of science

Scientific models

Rutherford's atomic model is a scientific model. Note the following about the nature of scientific models.

- Scientific models are proposed by scientists to **explain** observations. They may **not actually be consistent with reality.**
- Although scientists cannot see the actual structure of an atom, they make **inferences** of its structure based on observations, using **imagination and creativity.**
- A model is often modified or replaced by a new one when it fails to explain new observations.

Rutherford's model nicely explains the results of the Geiger-Marsden experiment (Fig 2.1c):

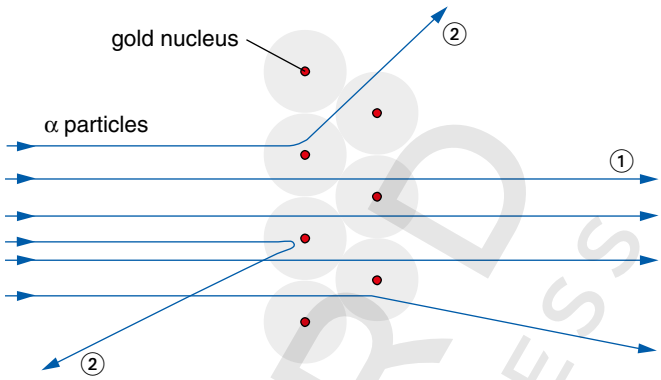


Fig 2.1c *Scattering of α particles explained.*

① Most α particles pass straight through the empty space of the gold atoms. Even if an α particle collides with an electron, it will only have negligible change in its path as its mass is much larger than that of an electron.

The mass of an α particle is about 7300 times that of an electron. Can you explain why α particles should not be replaced by β particles in this experiment?
A β particle will deflect significantly when it collides with an electron.

② A few of the α particles which happen to come close to the nuclei are repelled by the strong electric force. They are deflected or even bounced back by the much heavier nuclei. The closer an α particle to the nucleus, the stronger the repulsive electric force, and the larger the deflection.

The mass of a gold nucleus is about 50 times that of an α particle.

Experiment 2a

Analogue of α scattering (p.141)

Video

→ This video demonstrates Expt 2a.

We may use the α scattering analogue apparatus (Fig 2.1d) to simulate the deflection of α particles by a gold nucleus. In this model, the ball represents an α particle and the hill represents a gold atom. The ramp is placed at different positions and the ball is allowed to roll down towards the hill. If the ball is closer to the centre of the hill (representing the gold nucleus), it will be deflected more (Fig 2.1e).

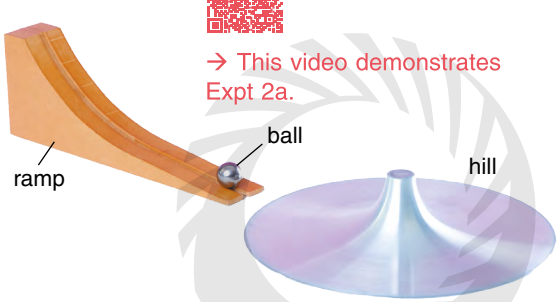


Fig 2.1d *α scattering analogue apparatus.*

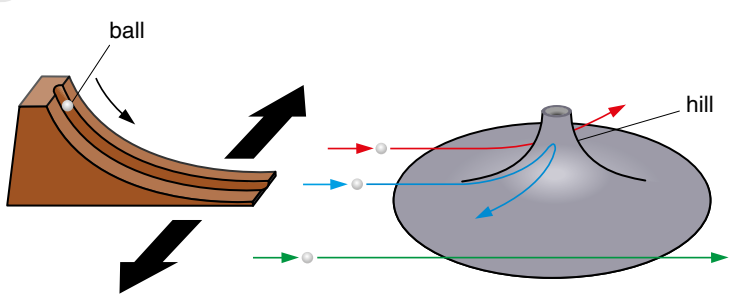


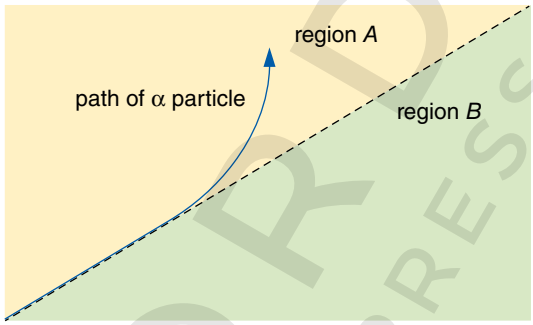
Fig 2.1e *Slope of hill causing ball to deflect.*

This set-up also simulates the change in energy of an α particle. When the ball rolls up the hill and gets closer to the centre, its kinetic energy is converted into its gravitational potential energy. Similarly, when the α particle gets closer to the gold nucleus, its kinetic energy is converted into the electric potential energy between it and the gold nucleus.

☆ HKDSE 2016 Paper 2 Q2.1, 2021 Paper 2 Q2.1

Example 1 | Path of an α particle

An α particle is directed to a thin gold foil. It is deflected by a gold nucleus and moves along the path as shown. The position of the gold nucleus is not shown in the figure.



Which of the following statements is/are correct?

(1) The gold nucleus is in region *A*.

(2) The change in momentum of the α particle will be very small if it collides with an electron, instead of the gold nucleus.

(3) The KE of the α particle is at a minimum when it is closest to the gold nucleus.

A (1) only

B (2) only

C (1) and (3) only

D (2) and (3) only

Solution

From the path, it can be inferred that the repulsive electric force acting on the α particle points towards region *A*. Therefore, the gold nucleus should be in region *B*.

∴ (1) is incorrect.

The mass of an α particle is much larger than that of an electron, so its change in momentum is very small when colliding with an electron.

∴ (2) is correct.

When the α particle approaches the gold nucleus, its KE is converted into the electric potential energy between it and the nucleus. When it is closest to the gold nucleus, the electric potential energy is at a maximum and the KE is at a minimum.

∴ (3) is correct.

∴ The answer is D.

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 *True or false:* In Geiger-Marsden experiment, most α particles pass straight through the empty space far from the nuclei. (T/F)

2 An α particle moves towards a gold nucleus (Fig a). It is closest to the nucleus at point *S*.

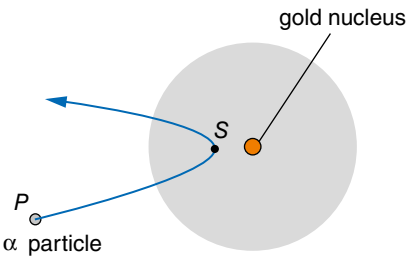


Fig a

Determine whether the following statements are true.

(a) At point *S*, the kinetic energy of the α particle is at a minimum. (T/F)

(b) The total energy of the α particle and the nucleus is lower when the particle is at point *S* than at point *P* (T/F)

(c) If the α particle has a larger initial kinetic energy, it can get closer to the nucleus. (T/F)

3 Determine whether the following statements about Rutherford's atomic model are true.

(a) All the positive charge of an atom is concentrated in the nucleus. (T/F)

(b) All the mass of an atom is concentrated in the nucleus. (T/F)

(c) There is no force acting on an electron by the nucleus. (T/F)

(d) Electrons distribute like a cloud all over an atom. (T/F)

4 Some α particles with the same initial speed are directed towards a gold foil. An α particle is deflected by a gold nucleus along path *X* (Fig b). Which path, *P*, *Q*, or *R*, is a possible path for another α particle?

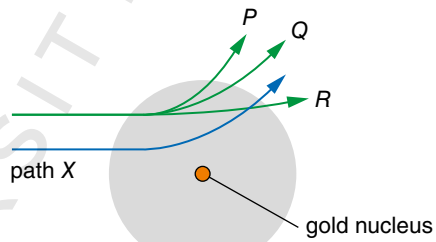


Fig b

DSE goal

State the limitations of Rutherford's atomic model in accounting for the motion of electrons around the nucleus and line spectra.

DSE exam

12(2)Q2.1, 13(2)Q2.2, 14(2)Q2.1, 21(2)Q.2(a)

3 Limitations of Rutherford's model

Rutherford's atomic model explained the results of the Geiger-Marsden experiment. However, it soon encountered some serious difficulties.

According to the theory of electromagnetism, a charge emits electromagnetic waves when it accelerates. Since the electrons in Rutherford's model had a centripetal acceleration, they were expected to emit EM waves continuously (Fig 2.1f).

Simulation

→ This simulation introduces Rutherford's atomic model and the limitations of the model. It is available on OUP web in PC version (.exe) only.

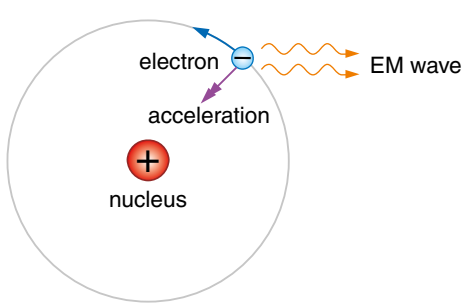


Fig 2.1f *An orbiting electron should emit EM waves continuously according to the theory of electromagnetism.*

According to classical electromagnetic theory, a changing E-field will produce a changing B-field and vice versa. This results in a self-sustaining EM wave and energy is transmitted out from the atom.

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This leads to the following two predictions, which are **inconsistent** with actual observations.

1 All atoms were expected to be unstable.

Since EM waves carry energy, an orbiting electron continuously emitting EM waves would lose its energy and spiral inwards (Fig 2.1g). The electron would soon crash into the nucleus and the atom would collapse. However, this prediction does not happen. Atoms are stable and do not collapse.

✎ When EM waves are emitted, the total energy of the orbital motion decreases. The orbital radius decreases too.

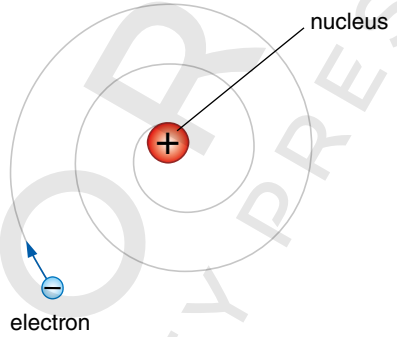


Fig 2.1g *In Rutherford's model, an electron would spiral inwards.*

2 The electrons in atoms were expected to emit EM waves with a continuous range of frequency.

The theory of electromagnetism says that the frequency of the emitted EM wave is equal to the orbital frequency of the electron. In Rutherford's model, the orbital frequency of an electron could take any value. Therefore, the waves emitted were expected to have a continuous range of frequency. However, as you will see in Chapter 2.2, atoms of a gas emit EM waves of certain **discrete** frequencies only.

✎ The proposing of new atomic model by Bohr serves a good scenario to illustrate an aspect of the nature of science: 'a model will be revised or replaced by a new one when it fails to explain new observations'.

These difficulties in explaining observations show that Rutherford's model should be modified. We will discuss this modified model in detail very soon.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Which of the following results **cannot** be explained by Rutherford's atomic model? Put a tick in the corresponding box(es).

- ☐ In Geiger-Marsden experiment, only a few α particles are deflected by large angles or bounced back.
- ☐ Atoms of gas emit EM waves of certain discrete frequencies only.
- ☐ Atoms are stable and do not collapse.
- ☐ Atoms are neutral in charge.

DSE goal

Realize the importance of scattering experiments in the discovery of the structure of atoms and the impact on the searching for new particles.

4 Importance of scattering experiments

The scattering of α particles by gold atoms guided Rutherford to propose his atomic model. Since then, scattering experiments have become a major tool for studying the sub-atomic world.

a Finding the structure of atoms and nuclei

Scattering experiments have played an important role in studying the structure of atoms and nuclei. Different types of particles are used to hit a target (which can be atoms or other particles). By studying the particles produced after bombardment, we can know more about the target.

b Searching for new particles

Sometimes, extra particles (other than those inside the apparatus before the bombardments) come out after the bombardments. Many particles, such as the proton and the neutron, were discovered in this way. *Quark*, an elementary particle, was also produced using a different version of scattering experiment.

An elementary particle is a particle not composed of other particles. Protons and neutrons are each composed of three quarks and therefore are not elementary particles.

Know more

Particle accelerators and new particles

Scientists have discovered many new particles from the collisions of high-speed particles using particle accelerators. Do you know how large a particle accelerator can be?

▸ p.93

Discovery of new particles

In 1918, Rutherford discovered the proton by bombarding nitrogen gas with α particles.

In 1932, James Chadwick showed that neutrons are emitted when α particles collide with beryllium. He was awarded the Nobel Prize in Physics in 1935 for this.

In 2012, scientists at the *Large Hadron Collider* discovered a new particle which was later confirmed to be the long-sought *Higgs boson* as predicted by the theory proposed by *François Englert* and *Peter Higgs*. Englert and Higgs were jointly awarded the Nobel Prize in Physics in 2013 for their theoretical contribution to the understanding of the origin of mass of sub-atomic particles.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 What are the uses of scattering experiments?

A Searching for new particles

B Finding the structure of atoms

C Both of the above

D None of the above

quark 夸克　Large Hadron Collider 大型強子對撞機　Higgs boson 希格斯玻色子
François Englert 恩格勒　Peter Higgs 希格斯

Exercise 2.1

1 Which of the following is **not** indicated by the results of the Geiger-Marsden experiment?

A The nucleus is positively charged.

B The nucleus contains most of the mass of the atom.

C The nucleus consists of protons and neutrons.

D The nucleus is extremely small in size compared to the whole atom.

***2** Which of the following predictions of Rutherford's atomic model is/are **inconsistent** with reality?

(1) Atoms would emit electromagnetic waves with a continuous range of wavelength.

(2) Electrons in atoms would lose energy continuously.

(3) Electrons in atoms would spiral inwards.

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

***3** Some α particles are directed towards a gold nucleus (Fig a). Which of the following are their possible paths?

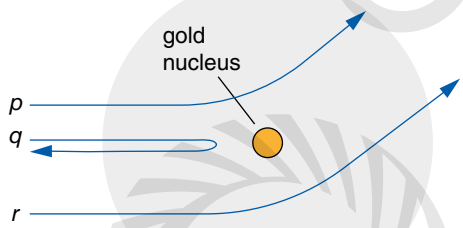


Fig a

A *p* and *q* only

B *p* and *r* only

C *q* and *r* only

D *p*, *q* and *r*

***4** (a) Explain how Rutherford's model suggested that all atoms were expected to be unstable.

(b) How the implication in (a) contradicts the actual observation?

***5** Explain the effect of electric forces on the particle in each of the following situations.

(a) An electron orbiting the nucleus of an atom in Rutherford's atomic model

(b) An α particle projected towards a gold nucleus

***6** Figure b shows the set-up of the Geiger-Marsden experiment. A beam of α particles is ejected from the radioactive source towards a thin gold foil. A detector is used to detect the α particles at different positions.

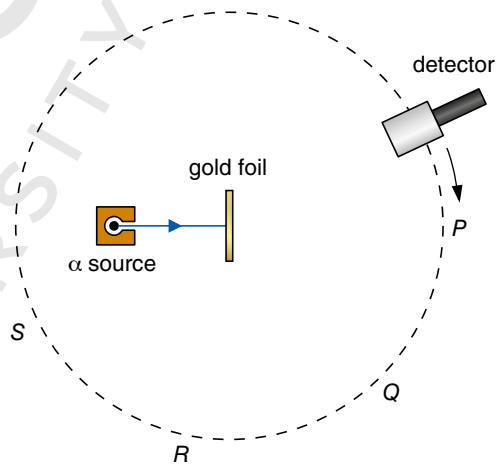


Fig b

(a) Explain why the experiment has to be performed in a vacuum.

(b) At which position, *P*, *Q*, *R* or *S*, is the largest number of α particles detected?

(c) How did Rutherford explain the result of this experiment?

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2.2 Atomic spectra and Bohr's model

→ This video demonstrates the Hands-on activity.

Video

Flipped classroom

Watch a video and answer the questions.

Hands-on activity

Producing a spectrum of light

What you need



What to do

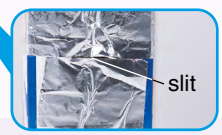


2 Make an opening above the DVD.

1 Make a slot at about 30° with the horizontal and slide the DVD into it with the shiny surface up.



3 Make an opening on the side opposite to the DVD. Fix two pieces of aluminium foil with clean edges close to each other. A narrow slit is formed.





4 Point the slit towards a light source and look at the DVD through the opening of the box.

What you find

What can you observe?

In this activity, the set-up works like a diffraction grating to produce a spectrum of light (Fig 2.2a). A clear spectrum allows us to work out the frequencies of the light that a light source consists of.

Cross-link

You learned grating in Book 3B Ch 6.3. ▶

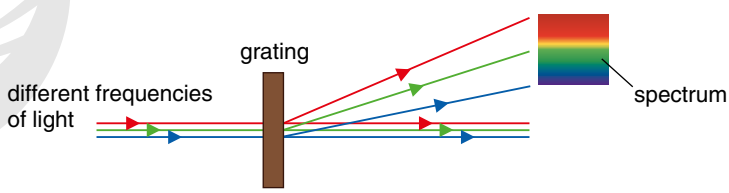


Fig 2.2a *How a grating forms a spectrum.*

A gas emits light in certain conditions and its spectrum can be formed in a similar way. In this part, we will study the spectra of gases and how *Niels Bohr* modified Rutherford's atomic model to explain the spectra.

Niels Bohr 玻爾

DSE goal

Describe the special features of line spectra of hydrogen atoms and other monatomic gases.

DSE exam

13(2)Q2.3

Experiment 2b

Emission spectrum (p.143)

Video

→ This video demonstrates Expt 2b.

1 Line spectrum

a Emission spectrum

The spectra of the light emitted by different light sources look different (Table 2.2a). Since the spectra are formed by the light emitted by the sources, they are called **emission spectra**.

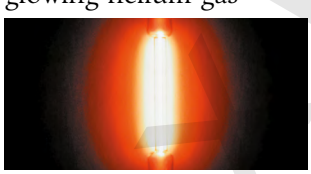
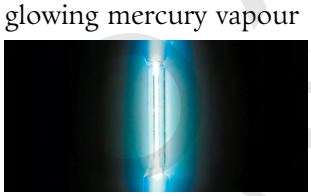
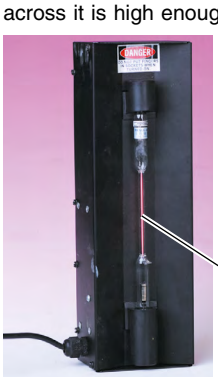
Light source	Spectrum
filament lamp (incandescent lamp) 	
glowing hydrogen gas 	
glowing helium gas 	
glowing mercury vapour 	

Table 2.2a *Spectra of light emitted by different sources.*

Each gas is sealed at low pressure in a gas discharge tube. It becomes hot and glows when the voltage applied across it is high enough.



The spectra of some other elements can be found in this website:

Objects like a hot glowing metal and a burning candle also produce continuous spectra.

The filament lamp produces a continuous spectrum. However, the gases produce **line spectra**, which consist of discrete bright lines (emission lines). Each line corresponds to a certain frequency. In other words, the gas atoms only emit lights of certain frequencies only.

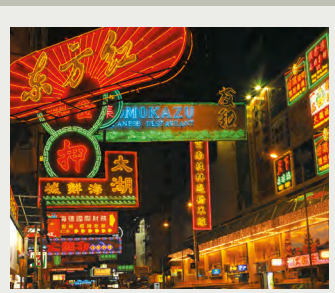
In general, a hot gas at low pressure produces a line spectrum, while a hot solid produces a continuous spectrum.

Know more

Neon signs

Neon signs can still be found in some streets like Tung Choi Street and Lockhart Road.

In the past, neon signs were commonly seen in Hong Kong. The signs are made using glass tubes containing gases like neon, helium and carbon dioxide. When electricity passes through the gases, each of the gases glows and emits light in its own emission spectrum. All the lights emitted in the tube combine to give the colour we see.



emission spectrum (pl. spectra) 發射光譜　　line spectrum 線狀光譜

DSE exam

12(2)Q2.2, 13(2)Q2.6, 14(2)Q2.2, 17(2)Q2.4, 19(2)Q2.1, 20(2)Q2.2

Experiment 2c

Absorption spectrum (p.146)

Video

→ This video demonstrates Expt 2c.

b Absorption spectrum

When a filament lamp is viewed through a low-density gas much colder than the filament, a continuous spectrum with a number of dark lines on it is observed. The dark lines suggest that light of certain frequencies are absorbed by the gas. This kind of spectrum is called the **absorption spectrum** (Fig 2.2b). The dark lines are called absorption lines.



Fig 2.2b An absorption spectrum of iodine vapour.

When comparing the emission spectrum and the absorption spectrum of the same element, it is found that the emission lines and the absorption lines have the same frequencies (Fig 2.2c).

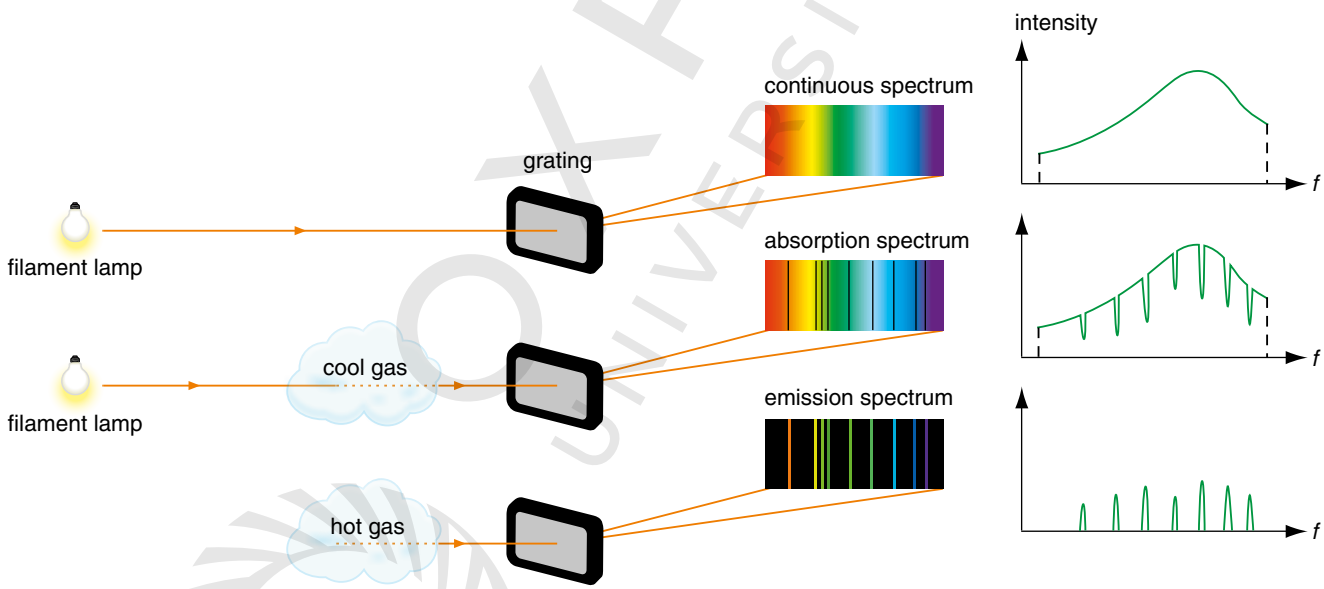
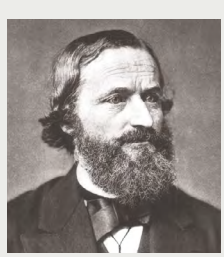


Fig 2.2c Absorption lines and emission lines of the same element have the same frequencies.

Know more

Gustav Kirchhoff (1824–1887)

Gustav Kirchhoff formulated the *Three Laws of Spectroscopy*, which describe different conditions for the production of and the appearance of continuous, emission and absorption spectra.



absorption spectrum 吸收光譜　Gustav Kirchhoff 基爾霍夫
Three Laws of Spectroscopy 光譜分析三大定律

It is like studying fingerprints to identify a person. ▶

In fact, each element has its unique spectrum. By studying the spectra of an unknown sample, we can determine what elements the sample contains. For example, the Sun's atmosphere absorbs light of certain frequencies from sunlight, forming absorption lines (Fig 2.2d). The composition of the atmosphere can be found based on these lines.



Fig 2.2d *Spectrum of sunlight, with absorption lines due to the Sun's atmosphere.*

DSE goal

Realize that energy of a hydrogen atom can only take on certain values.

Realize line spectra as evidence of energy levels of atoms.

DSE exam

15(2)Q2.2, 18(2)Q2.3

C Evidence of the energy levels of atoms

According to Einstein's photon model, the frequency of light is related to the energy of photons. The line spectra of hydrogen and some *monatomic gases* show that the atoms can emit and absorb energy of certain discrete values only. This gives evidence to the suggestion that the energy of an atom cannot change continuously but jumps from one level to another (Fig 2.2e). That is, an atom has discrete **energy levels**.

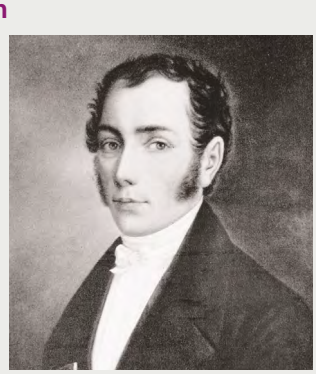


Fig 2.2e *The energy of an atom cannot change continuously like the PE of an object on a ramp, but jumps from one level to another like the PE of an object on a staircase.*

Know more

Discovery of the absorption spectrum of the Sun

The absorption spectrum of the Sun was discovered as early as 1802, but it was *Joseph von Fraunhofer* who studied the absorption lines carefully, and so the spectral lines are now called *Fraunhofer lines*.



Joseph von Fraunhofer (1787–1826).

monatomic gas 單原子氣體 energy level 能級 Joseph von Fraunhofer 夫瑯和費 Fraunhofer lines 夫瑯和費線

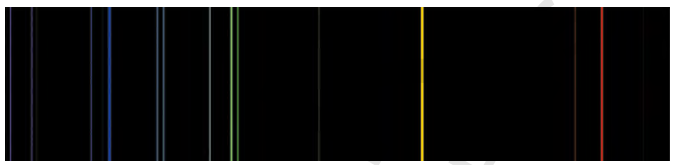
Example 2 | Line spectra

The emission spectra for hydrogen, helium and carbon are shown below.

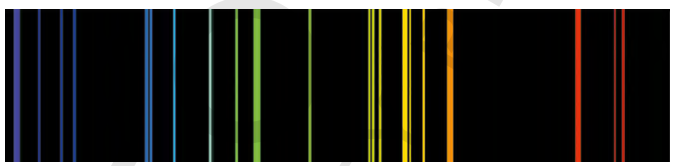
Hydrogen:



Helium:

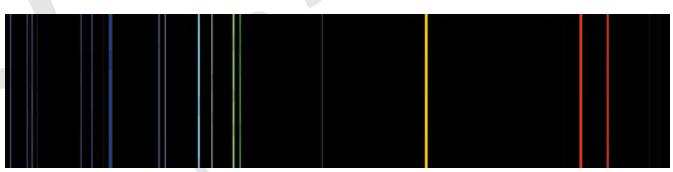


Carbon:

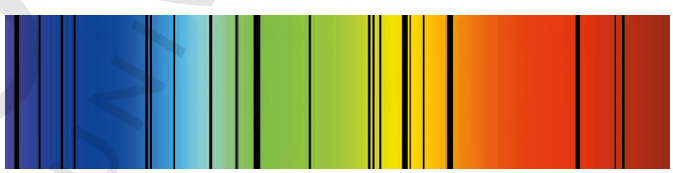


The emission spectrum of an unknown sample *A* and the absorption spectrum of another unknown sample *B* are shown below.

Sample *A*:



Sample *B*:



(a) Determine the element composition of samples *A* and *B*.

(b) What evidence about the energy of an atom does a line spectrum provide?

Solution

(a) Sample *A* is composed of hydrogen and helium as the spectral lines of these elements can be found in the emission spectrum of sample *A*.

Sample *B* is composed of carbon as the spectral lines of this element can be found in the absorption spectrum of sample *B*.

(b) The energy of an atom can only take certain discrete values.

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Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 Which of the following can produce a continuous spectrum? Put a tick in the corresponding box(es).

- ☐ A filament lamp
- ☐ A steel spoon heated to glow
- ☐ Hot gas at low pressure

2 *True or false:*

(a) The line spectrum of hydrogen gas is evidence that atoms have discrete energy levels. (T/F)

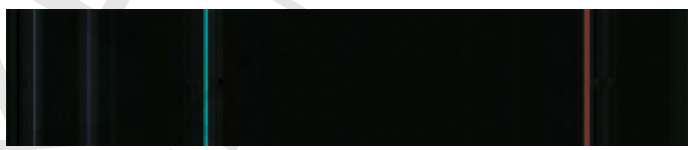
(b) The emission lines and absorption lines of the same element have the same frequencies. (T/F)

(c) The absorption spectrum of the Sun can be observed and analysed to determine the chemical composition of its atmosphere. (T/F)

(d) There are dark lines in the spectrum of sunlight because the Sun's atmosphere absorbs light of certain frequencies. (T/F)

3 The emission spectra for hydrogen, helium and lithium are as shown (Fig a). Figure b shows the spectra of samples *A* and *B*.

Hydrogen:



Helium:



Lithium:



Fig a

Sample *A*:



Sample *B*:



Fig b

(a) Are the spectra of *A* and *B* emission spectra or absorption spectra?

(b) Find the element composition of *A* and *B*.

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2 Bohr's model of the hydrogen atom

As discussed in Chapter 2.1, Rutherford's atomic model predicted that an atom could emit EM waves of any frequency. This contradicts with the line spectra of gases. To overcome the limitations of Rutherford's model, Niels Bohr proposed his atomic model, known as **Bohr's model**, of the hydrogen atom in 1913.

A hydrogen atom consists of an electron and a proton. It is the simplest element.

DSE goal

State the postulates of Bohr's model.

Distinguish between the quantum and classical aspects in the postulates.

Realize the postulate $m_e vr = nh/2\pi$ as the quantization of angular momentum of an electron around a hydrogen nucleus.

DSE exam

18(2)Q2.2

a Bohr's postulates

Bohr proposed several postulates when building his model. Some of them are just idea assumed to be true and are not based on well-established theory. However, they help explain the observations.

i Classical aspect

Postulate 1: Similar to Rutherford's model, the electron in a hydrogen atom moves around the nucleus in a circular orbit (Fig 2.2f). The centripetal force is provided by the electric force acting on the electron by the nucleus.

Cross-link

Review Book 2 Ch 9.2 for the centripetal force and Book 4 Ch 1.2 for the electric force.

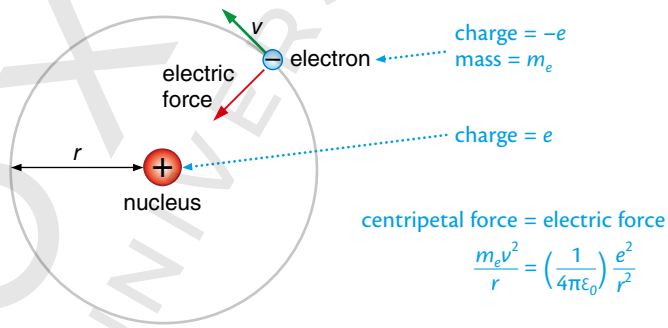


Fig 2.2f *The electron revolves around the nucleus in a circular orbit.*

ii Quantum aspects

Postulate 2: The electron can only occupy certain discrete orbits, each corresponding to an energy level (Fig 2.2g). Each orbit is specified by an integral number n, known as the **quantum number**.

These orbits are called stationary orbits. The electron cannot have any other orbit between these orbits.

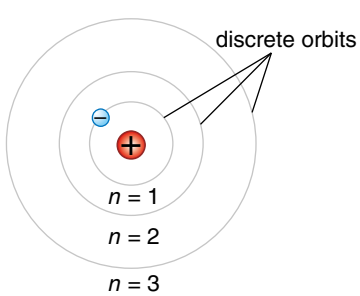


Fig 2.2g *Discrete orbits of the electron.*

Bohr's model 玻爾模型　　quantum number 量子數

See limitation 1 of Rutherford's model on p.47. With this postulate, an electron can remain in a stable orbit without spiralling inwards and crashing into the nucleus.

Postulate 3: As long as the electron stays in an orbit, it does not emit radiation even though it is accelerating. Its total energy remains constant.

Postulate 4: The electron can jump from one allowed orbit to another by gaining or losing energy. This can be done by absorbing or emitting a photon (Fig 2.2h).

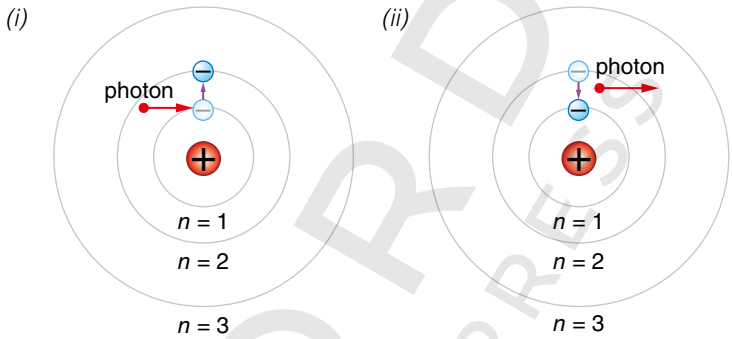


Fig 2.2h *Jumping of electron from one allowed orbit to another by (i) absorbing or (ii) emitting a photon.*

Postulate 5: The **angular momentum** of the electron is quantized in integral multiples of $\frac{h}{2\pi}$ (or $\hbar$).

Consider an electron of mass m_e moving at speed v in a circular orbit of radius r (Fig 2.2i). The magnitude of the angular momentum L of the electron is defined to be the product $r \times p$ where p is the linear momentum of the electron.

$$L = r \times p = r \times m_e v = m_e v r$$

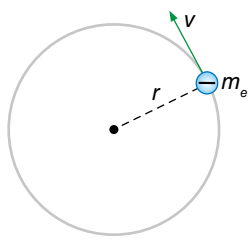


Fig 2.2i *Quantities involved in the calculation of angular momentum L.*

For an electron in the orbit of quantum number n, Bohr's postulate requires that

$$L_n = n\left(\frac{h}{2\pi}\right)$$

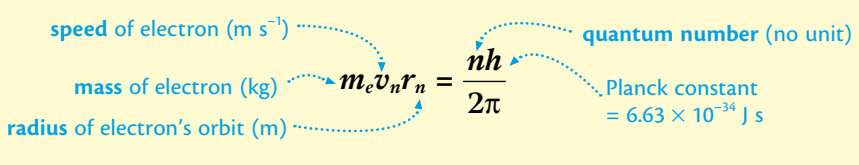
Know more

Angular momentum

In Book 2 Ch 7, we studied (linear) momentum $p = mv$ for linear motion. When net force $F = 0$, p of the object is conserved.

Angular momentum L is an equivalent physical quantity for circular motion. When net moment $\tau = rF = 0$, $L = rp$ of the object is conserved.

Moment was introduced in Book 2 Ch 5.



angular momentum 角動量

Know more

Niels Bohr (1885–1962)



Bohr was awarded the Nobel Prize in Physics in 1922 for his investigation of the structure of atoms and the radiation emitted from them.

OUP eResources

The postulates of Bohr's model of the hydrogen atom:

Classical aspect

1 The electron is a particle moving around the nucleus in a circular orbit. The centripetal force is provided by the electric force.

Quantum aspects

2 The electron can only occupy certain discrete orbits.

3 When the electron stays in an orbit, it does not radiate and its total energy remains unchanged.

4 The electron can jump from one orbit to another by gaining or losing energy. This can be done by absorbing or emitting a photon.

5 The angular momentum of the electron is quantized in integral multiples of $\frac{h}{2\pi}$.

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Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 Are the following the postulates of Bohr's model of the hydrogen atom? Do they show the quantum aspect of the model? Put a tick in the corresponding boxes.

		Postulate of Bohr's model?	Quantum aspect?
(a)	An atom is a large positively charged sphere with electrons placed all over it.		
(b)	The electron moves around the nucleus in a circular orbit.		
(c)	The total energy of an electron remains unchanged when it stays in an orbit.		
(d)	The angular momentum of the electron is quantized.		
(e)	The electron can only stay in some discrete orbits.		
(f)	The electron can jump from one energy level to another by absorbing or emitting a photon.		

2 *True or false:* The postulates of Bohr's model are based on well-established theory. (T/F)

DSE exam
18(2)Q2.4

b Discrete orbits

With the postulates assumed by Bohr, the radius of the nth orbit of a hydrogen atom is found to be:

radius of nth orbit (m) $r_n = n^2 r_1$ **radius** of innermost (n = 1) orbit (m); **quantum number** (no unit)

r_1 is the radius of the n = 1 orbit. It is equal to 5.29×10^{-11} m. The larger the quantum number n, the larger the radius. The first three of these discrete orbits are drawn to scale in Figure 2.2j. Note that the difference in radius between adjacent orbits increases as n increases.

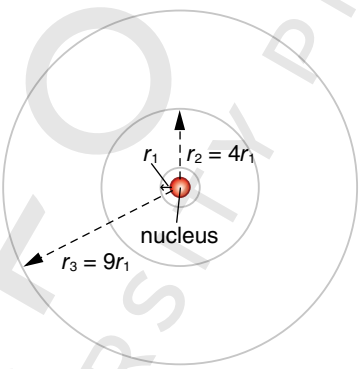


Fig 2.2j *Discrete electron orbits in Bohr's model.*

Know more

Derivation of $r_n = n^2 r_1$

From Figure 2.2f (p.56), when the orbital radius is r_n,

$$\frac{m_e v_n^2}{r_n} = \left(\frac{1}{4\pi\varepsilon_0}\right)\frac{e^2}{r_n^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

By quantization of angular momentum of electron, $m_e v_n r_n = \frac{nh}{2\pi}$

$$v_n = \frac{nh}{2\pi m_e r_n} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Substitute (2) into (1),

$$\frac{m_e}{r_n}\left(\frac{nh}{2\pi m_e r_n}\right)^2 = \left(\frac{1}{4\pi\varepsilon_0}\right)\frac{e^2}{r_n^2}$$

$$r_n = \left(\frac{\varepsilon_0 h^2}{\pi m_e e^2}\right) n^2 \quad \ldots\ldots\ldots\ldots (3)$$

The radius of the innermost orbit (n = 1) is

$$r_1 = \left(\frac{\varepsilon_0 h^2}{\pi m_e e^2}\right)(1)^2 = \frac{\varepsilon_0 h^2}{\pi m_e e^2} \quad \ldots\ldots\ldots\ldots (4)$$

$$= 5.29 \times 10^{-11} \text{ m}$$

Substitute (4) into (3),

$$r_n = n^2 r_1$$

☆ HKDSE 2022 Paper 2 Q2.4

Example 3 Bohr orbit

According to Bohr's model, the radius of the innermost orbit of the electron in a hydrogen atom is 5.29×10^{-11} m.

(a) Find the radius of the electron's orbit corresponding to quantum number $n = 2$.

(b) If the electron moves to an orbit corresponding to a larger n, how will its speed change?

When n increases, the linear momentum p decreases but the angular momentum L increases.
$L = \frac{nh}{2\pi} \propto n$

Solution

(a) By $r_n = n^2 r_1$,

$r_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10}$ m

(b) By $m_e v_n r_n = \frac{nh}{2\pi}$ and $r_n = n^2 r_1$,

$$v_n = \frac{nh}{2\pi m_e r_n} = \frac{nh}{2\pi m_e (n^2 r_1)} = \frac{h}{2\pi m_e n r_1} \propto \frac{1}{n}$$

If n becomes larger, the speed v_n will become smaller.

Instead of using angular momentum, you may also solve this question by considering the electric force and centripetal force:
$\frac{e^2}{4\pi\varepsilon_0 r_n^2} = \frac{m_e v_n^2}{r_n}$

Caution
We cannot use $v_n = \frac{nh}{2\pi m_e r_n}$ directly and say that $v_n \propto n$ as r_n is not a constant.

This is similar to circular motion under gravity.

You can see that the electron becomes **slower** after moving to an **outer orbit**.

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Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 Complete the table based on Bohr's model of the hydrogen atom.

		r_n
	$n = 1$	5.29×10^{-11} m
(a)	$n = 4$	
(b)	$n = 5$	

2 Consider Bohr's model of the hydrogen atom. Determine whether the following statements are true.

(a) The electron has a larger angular momentum at the outer orbit. (T/F)

(b) The electron has a higher KE at the outer orbit. (T/F)

Learn by practice 1

1 Consider Bohr's model of the hydrogen atom. The electron is in an orbit corresponding to quantum number $n = 3$.

(a) Find the angular momentum of the electron in terms of h and π.

(b) Find the radius of the orbit in terms of r_1 (radius of the innermost orbit).

DSE goal

Realize the equation for the energy of an electron in a hydrogen atom as $E_n = -\frac{13.6\text{ eV}}{n^2}$.

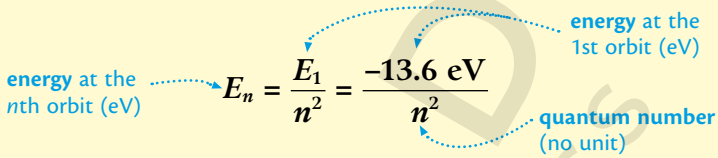
Use eV as a unit of energy.

DSE exam

12(2)Q2.5

C Discrete energy levels

In Bohr's model of the hydrogen atom, the total energy (KE + electric PE) possessed by the electron in the nth orbit is given by:



See *Know more* on p.62 for the derivation.

E_n is also given by $E_n = -\frac{1}{n^2}\left(\frac{m_e e^4}{8h^2\varepsilon_0^2}\right)$

Note that:

1. The electron energy is quantized. These discrete energy values form the **energy levels** of the hydrogen atom.

Sometimes 'energy levels of the electron' is used instead of 'energy levels of the atom' as they are equivalent.

2. Each energy level corresponds to an orbit of the electron. An electron in an orbit of radius r_n has a total energy of E_n.
3. The **negative sign** of energy means that the electron is **bound** by the nucleus. Work has to be done to remove the electron from the atom to infinity. When $n = \infty$, the total energy of the electron is zero ($E_\infty = 0$).

At infinity, the electron is not bound by the nucleus anymore, i.e. the electron is free. See p.66 for details.

4. The energy is lower (more negative) for lower energy level (smaller n). This means that energy is needed for an electron to jump from a lower level to a higher level.

When the electron is in an orbit with $\boldsymbol{n = 1}$, it has the lowest permitted energy (i.e. E_1). The atom is said to be in its **ground state**. When the electron is in an orbit with $\boldsymbol{n}$ **larger than 1**, the electron is at a higher energy level (E_2, E_3, etc.). The atom is said to be in an **excited state**.

Table 2.2b shows the energy levels of the hydrogen atom and the radii of the electron's orbits in the first four states.

State	Quantum number	Energy level	Radius of orbit
ground state	$n = 1$	$E_1 = -13.6$ eV	$R_1 = 5.29 \times 10^{-11}$ m
1st excited state	$n = 2$	$E_2 = -\frac{13.6}{2^2} = -3.40$ eV	$R_2 = (2)^2(5.29 \times 10^{-11}) = 2.12 \times 10^{-10}$ m
2nd excited state	$n = 3$	$E_3 = -\frac{13.6}{3^2} = -1.51$ eV	$R_3 = (3)^2(5.29 \times 10^{-11}) = 4.76 \times 10^{-10}$ m
3rd excited state	$n = 4$	$E_4 = -\frac{13.6}{4^2} = -0.85$ eV	$R_4 = (4)^2(5.29 \times 10^{-11}) = 8.46 \times 10^{-10}$ m

Table 2.2b *Energy levels and radii of orbits in different states.*

ground state 基態　　excited state 受激態

The energy levels of a hydrogen atom can be represented by an energy-level diagram (Fig 2.2k). As n increases, the difference in energy between adjacent energy levels (i.e. $E_{n+1} - E_n$) **decreases**.

Discrete energy levels also exist in the atoms of other elements, but the calculation of the energy levels is more complicated.

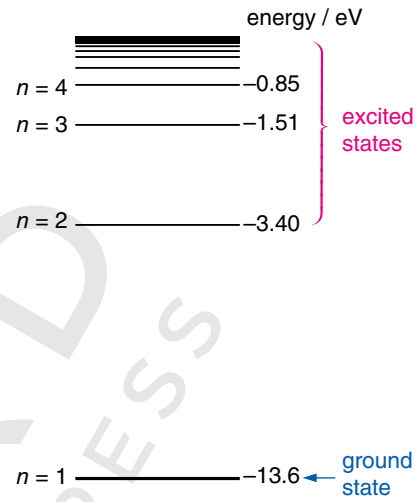


Fig 2.2k *Energy-level diagram for a hydrogen atom (not drawn to scale).*

Know more

Derivation of $E_n = \frac{E_1}{n^2} = -\frac{13.6\text{ eV}}{n^2}$ for a hydrogen atom

Recall equation (1) on p.59.

$$\frac{m_e v_n^2}{r_n} = \left(\frac{1}{4\pi\varepsilon_0}\right)\frac{e^2}{r_n^2}$$

$$\frac{1}{2}m_e v_n^2 = \left(\frac{1}{8\pi\varepsilon_0}\right)\frac{e^2}{r_n} = \left(\frac{1}{8\pi\varepsilon_0}\right)\frac{e^2}{n^2 r_1}$$

This gives the KE of the electron in the nth orbit.

Electric PE at nth orbit $= \frac{Q_1 Q_2}{4\pi\varepsilon_0 r_n} = \frac{(e)(-e)}{4\pi\varepsilon_0(n^2 r_1)} = \left(-\frac{1}{4\pi\varepsilon_0}\right)\frac{e^2}{n^2 r_1}$

The formula of electric potential energy of two point charges is new to Ss.

$$E_n = \left(\frac{1}{8\pi\varepsilon_0}\right)\frac{e^2}{n^2 r_1} + \left(-\frac{1}{4\pi\varepsilon_0}\right)\frac{e^2}{n^2 r_1} = \left(-\frac{e^2}{8\pi\varepsilon_0 r_1}\right)\frac{1}{n^2} \quad \ldots\ldots\ldots\ldots \quad (5)$$

$$E_1 = \left(-\frac{e^2}{8\pi\varepsilon_0 r_1}\right)\frac{1}{1^2} = -2.18 \times 10^{-18}\text{ J} = -13.6\text{ eV}$$

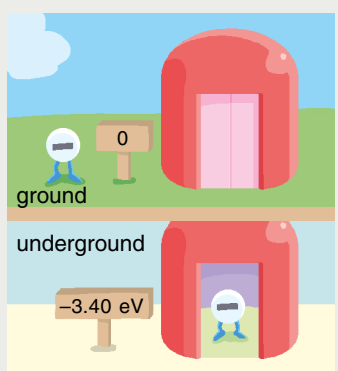
Substitute this into (5),

$$E_n = \frac{E_1}{n^2} = -\frac{13.6\text{ eV}}{n^2}$$

Negative sign of E

The negative sign of E can be understood by considering an underground railway station. If you want to leave the station from an underground floor, you need energy to go up to the ground floor. If the energy at the ground floor is taken as 0, your energy at the underground floor is negative.

Similarly, an electron bound to an atom needs energy to leave the atom ($r = \infty$). Since its energy at $r = \infty$ is taken as 0, its energy at any energy level should be negative.



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Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 Determine whether the following statements about Bohr's model are true.

(a) The total energy possessed by the electron orbiting the nucleus in a hydrogen atom can be positive or negative. (T/F)

(b) The equation $E_n = -\frac{13.6\text{ eV}}{n^2}$ only applies to a hydrogen atom. (T/F)

(c) A hydrogen atom is in the second excited state when its quantum number n is 2. (T/F)

(d) A hydrogen atom is at the energy level of 0 eV when it is in the ground state. (T/F)

2 Complete the table based on Bohr's model of the hydrogen atom.

	n	E_n / eV
	1	−13.6
(a)	3	
(b)		−0.544

DSE goal

Apply $E_n = -\frac{13.6\text{ eV}}{n^2}$ to solve problems.

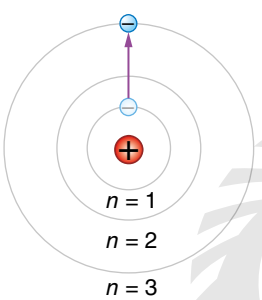
DSE exam

15(2)Q.2, 22(2)Q2.4

d Electron transition

The process of an electron in an atom jumping from one energy level to another is called **electron transition**. If an electron in the **ground state** gains the right amount of energy, called the **excitation energy**, it will jump to an excited state (Fig 2.2l). This can be done by absorbing a photon with energy exactly the same as the energy difference between the two states.

We can also represent an electron transition like this:



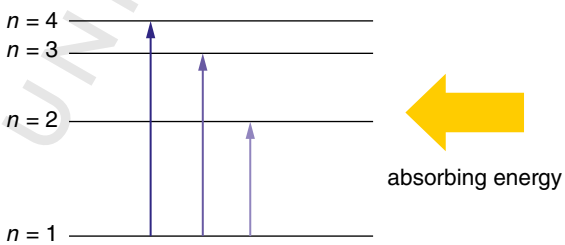


Fig 2.2l Some possible excitations.

The excitation energy is the exact amount of energy absorbed to raise an electron from the ground state to a certain excited state (i.e. 1 → n).

Excitation energy = $E_n - E_1$

In general, when an electron jumps from a lower energy level E_b to a higher energy level E_a, it **absorbs** energy.

Energy absorbed = $E_a - E_b$

electron transition 電子躍遷 excitation energy 激發能

An electron in an excited state is unstable. It will drop either directly to the ground state, or via some intermediate states before reaching the ground state (Fig 2.2m). The energy will be released as **a photon in each transition**.

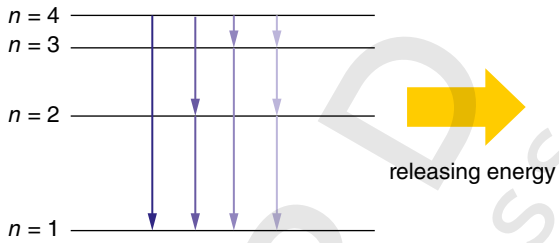


Fig 2.2m *Possible paths of an electron dropping from an excited state n = 4 to the ground state.*

When the electron at a higher energy level E_a drops to a lower energy level E_b, it **releases** energy.

Energy released = $E_a - E_b$

Example 4 Raising an electron to higher energy states

Consider Bohr's model of the hydrogen atom. The ground state energy of the electron is –13.6 eV.

(a) State the physical significance of energy levels having negative values.

(b) Find the excitation energy required to raise the electron from the ground state to the first excited state ($n = 2$).

(c) Can the electron in the ground state absorb a photon of energy 10.3 eV?

(d) Find the energy required for the electron to jump from the first excited state to the second excited state.

Solution

(a) The negative values of energy levels mean that the electron is bound by the nucleus.

(b) Excitation energy = $E_2 - E_1 = \left(\frac{-13.6}{2^2}\right) - (-13.6) = 10.2$ eV

(c) No

(d) Energy required = $E_3 - E_2 = \left(\frac{-13.6}{3^2}\right) - \left(\frac{-13.6}{2^2}\right) = 1.89$ eV

Tip

To raise an electron to a higher energy state, the energy of the photon must be exactly the same as the energy difference between the two states.

Think about...

Will the energy required for the electron to jump from the second excited state to the third excited state be larger or smaller than the answer in (d)?

Example 5 Possible electron transitions

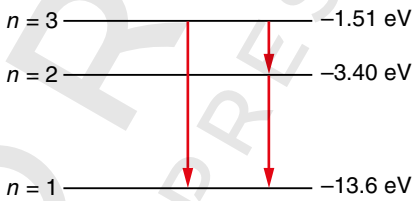
An electron in a hydrogen atom returns from the second excited state to the ground state. Draw a diagram to show all the possible transitions and find the energy released in each transition.

Given: $E_1 = -13.6$ eV, $E_2 = -3.40$ eV, $E_3 = -1.51$ eV

Solution

All possible transitions are as shown.

Transition	Energy released
$n = 3$ to $n = 1$	$(-1.51) - (-13.6) = 12.09$ eV
$n = 3$ to $n = 2$	$(-1.51) - (-3.40) = 1.89$ eV
$n = 2$ to $n = 1$	$(-3.40) - (-13.6) = 10.2$ eV



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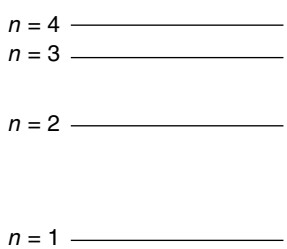
Quick check 8

→ E-version in OUP Exercise Platform is also available.

1 The ground state energy of the electron in a hydrogen atom is –13.6 eV. Complete the table.

	Electron transition	Is energy absorbed or released?	Energy change
(a)	$n = 1$ to $n = 4$		
(b)	$n = 4$ to $n = 3$		

2 The electron in a hydrogen atom returns to its ground state from the third excited state. How many possible transitions are there?



Learn by practice 2

The ground state energy of the electron in a hydrogen atom is –13.6 eV.

1 The electron in a hydrogen atom transits from the first excited state to the fourth excited state. Find the energy absorbed.

2 The electron in a hydrogen atom returns from state $n = 4$ to $n = 2$. Draw an energy level diagram with arrows to show all possible transitions, and find the energy released in each transition.

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DSE goal

Distinguish ionization and excitation energies.

e Ionization

If an electron in an atom gains enough energy, it can escape from the atom completely and becomes a free electron. The atom is said to be **ionized**.

The KE of the electron is zero after absorbing this minimum energy. ▶ The **minimum** energy required for an atom in its **ground state** to be ionized is called the **ionization energy**. The atom absorbing the ionization energy is raised to state $n = \infty$ with energy $E_\infty = 0$.

The ionization energy is the energy just sufficient to remove an electron from an atom in its ground state (i.e. $n = 1 \rightarrow \infty$).

Ionization energy = $E_\infty - E_1$

If an atom is in state n, the minimum energy required to ionize it is

$$E = E_\infty - E_n$$

Recall that an atom can jump to a higher energy level after absorbing a photon with an **exact** amount of energy equal to the difference between the two energy levels. However, ionizing an atom does not need such an exact amount of energy. If the energy needed to ionize an atom is E, the atom can absorb any photon with energy $\geq E$ (Fig 2.2n). The extra energy will become the KE of the electron and the nucleus (or ion).

The energy of a **free** electron is not quantized. ▶

The whole photon is absorbed. ▶

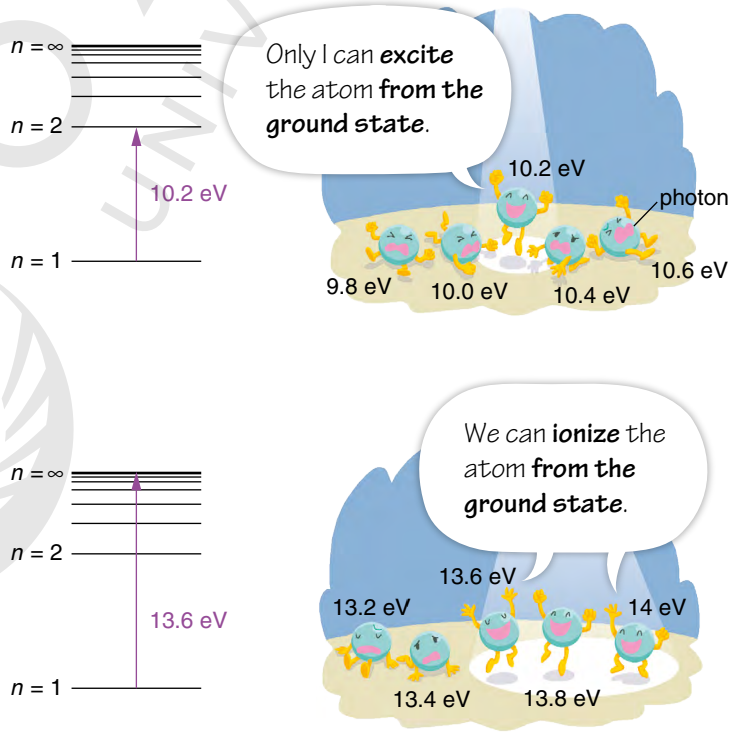


Fig 2.2n *Exciting and ionizing a hydrogen atom.*

ionization energy 電離能

Example 6 | Ionizing a hydrogen atom

The ground state energy of a hydrogen atom is –13.6 eV.

(a) Find the ionization energy of a hydrogen atom.

(b) A hydrogen atom is in an excited state. Is the minimum energy needed to ionize it higher or lower than the ionization energy?

(c) A hydrogen atom is in the first excited state. Can it absorb a photon of energy 10 eV? Explain.

Solution

(a) Ionization energy = $E_\infty - E_1 = 0 - (-13.6) = 13.6$ eV

(b) Lower

(c) Minimum energy to ionize the atom = $E_\infty - E_2 = 0 - \left(-\frac{13.6}{2^2}\right) = 3.4$ eV

Since the photon energy is higher than 3.4 eV, the atom can absorb the photon and become ionized.

Tip
An atom in an excited state has higher energy, so less energy is needed to ionize it.

The electron and the nucleus will gain extra KE after absorbing the photon.

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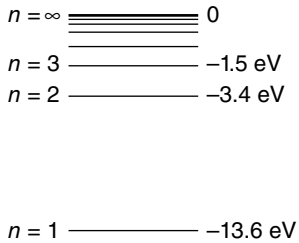
Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 The minimum energy needed to ionize a hydrogen atom in state n is E. The ground state energy of the atom is –13.6 eV. Complete the table.

	State	E / eV
(a)	$n = 1$	
(b)		0.85

2 Some energy levels of a hydrogen atom are as shown.



Suppose a photon of energy E falls on a hydrogen atom in the ground state.

(a) Can the atom be excited if

(i) $E = 10.2$ eV;

(ii) $E = 10.7$ eV?

(b) Can the atom be ionized if

(i) $E = 13.6$ eV;

(ii) $E = 13.9$ eV?

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DSE goal

Explain spectral lines in terms of difference in energies.

Derive the expression $\frac{1}{\lambda} = \frac{13.6\text{ eV}}{hc}\left(\frac{1}{b^2} - \frac{1}{a^2}\right)$ for the wavelength of photon emitted or absorbed when an electron undergoes a transition.

Apply this equation and $E = hf$ to solve problems.

Interpret line spectra using Bohr's equation of electron energy.

DSE exam

16(2)Q2.3, 17(2)Q.2, 19(2)Q2.5, 20(2)Q2.3–Q2.5, 21(2)Q2.5, Q.2, 22(2)Q2.3, 23(2)Q.2(b), etc.

3 Explaining spectra of hydrogen

a Emission spectrum

The emission lines of hydrogen can be explained by Bohr's atomic model and Einstein's photon model.

When an electron at a higher energy level E_a drops to a lower energy level E_b, it releases energy as a **single photon** (Fig 2.2o).

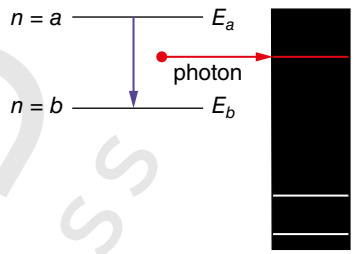


Fig 2.2o *Electron transition and emission of photon.*

Since a photon carries an energy hf and $c = f\lambda$,

frequency of light (Hz) ····· speed of light = 3×10^8 m s^{-1}

Planck constant = 6.63×10^{-34} J s ····· $hf = \frac{hc}{\lambda} = E_a - E_b$ ····· **energy** of lower level (J)

wavelength of light (m) ····· **energy** of higher level (J)

As the energy levels are quantized, the frequencies and wavelengths of the photons emitted can be some discrete values only. Therefore, discrete bright lines corresponding to these wavelengths are observed in the emission spectrum. These wavelengths can be further expressed as follows:

$$\frac{hc}{\lambda} = -\frac{13.6\text{ eV}}{a^2} - \left(-\frac{13.6\text{ eV}}{b^2}\right)$$

$$\frac{1}{\lambda} = \frac{13.6\text{ eV}}{hc}\left(\frac{1}{b^2} - \frac{1}{a^2}\right)$$

This equation can be applied to other elements. Each element has a unique set of energy levels and produces its own characteristic spectral lines.

Simulation

→ This simulation shows the Bohr's model to illustrate that EM waves of different wavelengths are emitted as the electron drops from a higher energy level to a lower energy level. It is available on OUP web in PC version (.exe) only.

This equation can be applied to hydrogen only.

b Absorption spectrum

When light of continuous frequency is sent through some hydrogen gas, most photons are unaffected and pass straight through the gas. However, a photon may be absorbed to raise an electron from a lower energy level E_b to a higher energy level E_a if that **single photon** has the **exact amount of energy** equal to $E_a - E_b$. The electron will then drop to a lower energy level and re-emit a photon **in random direction** (Fig 2.2p).

Simulation

→ This simulation explains the formation of the absorption spectrum. It is available on OUP web in PC version (.exe) only.

The excited electron may drop from E_a to lower energy levels via various paths, emitting photons corresponding to different wavelengths.

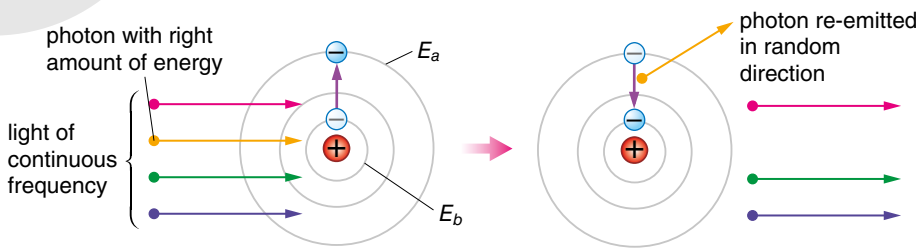


Fig 2.2p *An electron absorbs and re-emits a photon.*

Light of frequency corresponding to the energy difference ($E_a - E_b$) has a much smaller intensity in the original direction due to the absorptions by electrons (Fig 2.2q). This results in a dark line in the background continuous spectrum.

The dark lines are not completely dark. They appear dark when compared to the bright background continuous spectrum.

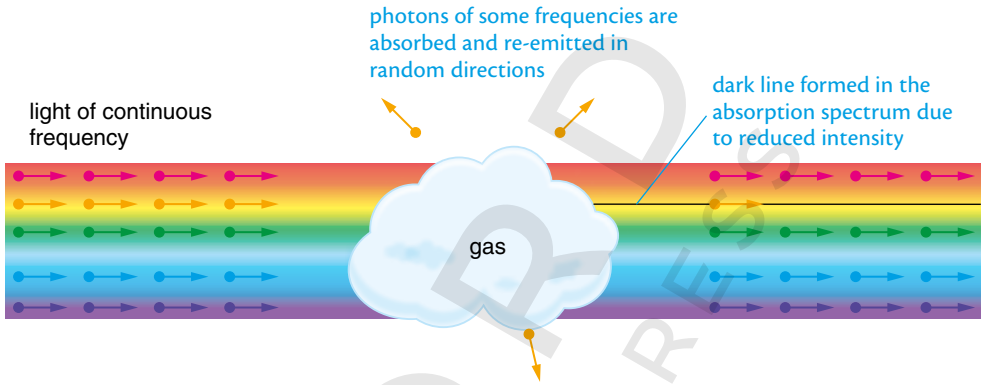


Fig 2.2q *How an absorption spectrum is formed.*

The formulae on p.68 also work in finding the frequencies and wavelengths of the absorption lines. The wavelengths of the photons absorbed can be some discrete values only. Therefore, discrete dark lines are observed in the absorption spectrum.

Know more

Different colours of fireworks

Do you know what gives the colours of fireworks?

▸ p.94

Example 7 Photons and electron transitions

A hydrogen atom absorbs a photon and is excited from the ground state to the third excited state ($n = 4$). Its ground state energy is –13.6 eV.

(a) Find the energy of the photon (in eV).

(b) The atom then drops from the third excited state to the first excited state ($n = 2$). Find the wavelength of the light emitted.

(c) The atom then drops from the first excited state to the ground state ($n = 1$). Find the frequency of the light emitted.

Solution

(a) $E = E_4 - E_1 = -\dfrac{13.6}{4^2} - (-13.6) = 12.75 \text{ eV}$

(b) $\dfrac{1}{\lambda} = \dfrac{13.6 \text{ eV}}{hc}\left(\dfrac{1}{b^2} - \dfrac{1}{a^2}\right) = \dfrac{(13.6)(1.6 \times 10^{-19})}{(6.63 \times 10^{-34})(3 \times 10^8)}\left(\dfrac{1}{2^2} - \dfrac{1}{4^2}\right)$

$\lambda = 4.88 \times 10^{-7} \text{ m}$

(c) By $hf = E_a - E_b$,

$(6.63 \times 10^{-34})f = \left[\dfrac{-13.6}{2^2} - (-13.6)\right] \times 1.6 \times 10^{-19}$

$f = 2.46 \times 10^{15} \text{ Hz}$

$\lambda = \dfrac{hc}{E} = 9.75 \times 10^{-8}$ m

This is UV radiation.

Tip

Express 13.6 eV in J.

This is cyan or blue light.

$\lambda = \dfrac{c}{f} = 1.22 \times 10^{-7}$ m

This is UV radiation.

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Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* If sunlight is sent through some hydrogen gas, light with certain frequencies will be completely absorbed. (T/F)

2 A hydrogen atom transits from one energy level to another. Complete the table about the EM radiation involved.

	Electron transition	λ / nm	f / 10^{14} Hz	Type of EM wave
(a)	$n = 5$ to $n = 2$			
(b)	$n = 3$ to $n = 6$			

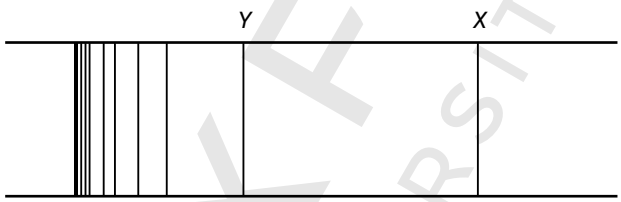
☆ HKDSE 2017 Paper 2 Q.2, 2020 Paper 2 Q2.3, 2021 Paper 2 Q.2

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Spectral lines of hydrogen

The figure below shows the emission lines of hydrogen corresponding to electron transitions from all excited states to the first excited state ($n = 2$).



(a) From which excited states do electrons drop from in producing lines X and Y? (2 marks)

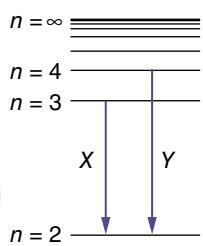
(b) Find the shortest wavelength of these lines. (2 marks)

(c) Can EM radiation of wavelength 360 nm be absorbed by hydrogen atoms in the first excited state? Explain. (3 marks)

Solution

(a) X: From second excited state ($n = 3$) 1A

Y: From third excited state ($n = 4$) 1A



(b) $\frac{1}{\lambda} = \frac{13.6\text{ eV}}{hc}\left(\frac{1}{b^2} - \frac{1}{a^2}\right)$

$= \frac{(13.6)(1.6 \times 10^{-19})}{(6.63 \times 10^{-34})(3 \times 10^{8})}\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$ 1M

$\lambda = 3.66 \times 10^{-7}$ m 1A

(c) Yes, 1A

the energy of a photon in the EM radiation is higher than the energy needed to ionize an atom, 1A

so some atoms will absorb the photons and become ionized. 1A

Common mistake
Students may not realize that since X is furthest away from another line, it is formed when the electron drops from the lowest energy level.

Common mistake
Students may not realize that the line of the shortest wavelength is due to the electron transition from $n = \infty$ to $n = 2$.

Common mistake
Students may not know that light with λ shorter than the answer in (b) can ionize the atom. Some confuse ionization with excitation and wrongly think that the photons cannot be absorbed.

▶ Revision 2 Q32 (p.92)

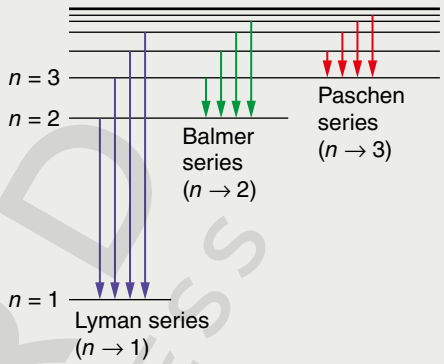
Know more

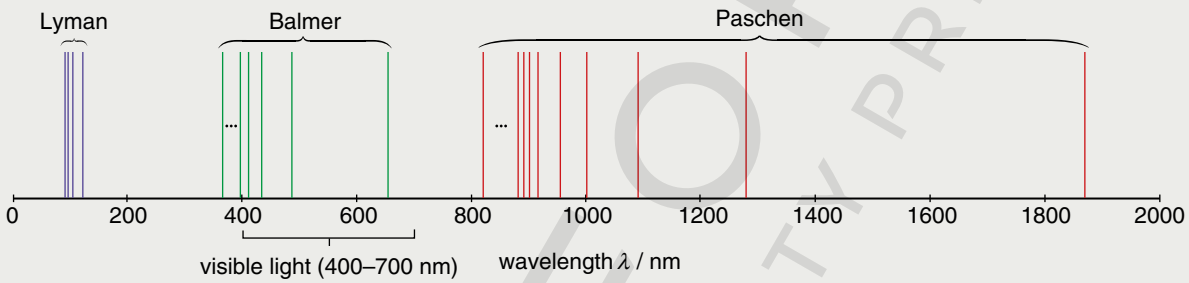
Spectral series and the Balmer formula

The line spectrum of hydrogen can be separated into different spectral series corresponding to different final states of the electron. Among them, the Balmer series ($n \rightarrow 2$) contains all four visible lines in the emission spectrum. The wavelengths of the lines in this series are found to obey the Balmer formula, which was discovered by *Johann Balmer* in 1885:

$$\frac{1}{\lambda} = R\left(\frac{1}{2^2} - \frac{1}{n^2}\right)$$

where R is a constant and is taken as $1.097 \times 10^7\ m^{-1}$





DSE exam

14(2)Q2.4, 15(2)Q2.6, Q.2(d)

When a substance is heated, its atoms can also jump to higher energy levels.

Cross-link

You learned elastic collisions and inelastic collisions in Book 2 Ch 7.1.

4 Transitions by collision

Besides absorbing a photon, an atom can also jump to a higher energy level when colliding with a particle (e.g. an electron outside the atom). **The particle is not absorbed** but only transfers some of its KE to the atom. The amount of KE transferred equals the difference between two energy levels. The total KE of the particle and the atom after the collision is smaller than that before, so the collision is **inelastic** (Fig 2.2r).

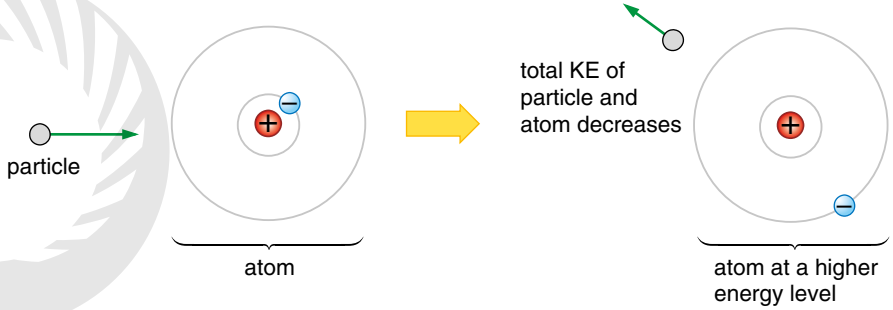


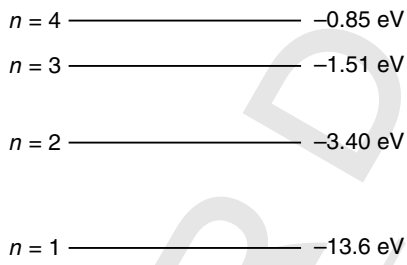
Fig 2.2r *An atom can jump to a higher energy level when collided by another particle.*

On the other hand, if the KE of the particle is **less than** the amount needed to raise the atom to a higher energy level, the particle will **not** transfer any of its KE to the atom. The total KE of the particle and the atom will be conserved. Therefore, the collision will be **elastic**.

Johann Balmer 巴耳末

Example 8 Excitation by electron and photon

The figure shows some energy levels of a hydrogen atom.



Determine whether a hydrogen atom in the ground state can be excited by each of the following particles, and to which state it can be excited to.

(a) An electron of kinetic energy 10.0 eV

(b) An electron of kinetic energy 12.2 eV

(c) A photon of energy 12.2 eV

Solution

From $n = 1$ to $n = 2$, $\Delta E = -3.40 - (-13.6) = 10.2$ eV

From $n = 1$ to $n = 3$, $\Delta E = -1.51 - (-13.6) = 12.09$ eV

From $n = 1$ to $n = 4$, $\Delta E = -0.85 - (-13.6) = 12.75$ eV

(a) As 10.0 eV is smaller than 10.2 eV, the atom cannot be excited.

(b) As 12.2 eV is larger than 10.2 eV and 12.09 eV, the atom can be excited to either the first excited state ($n = 2$) or the second excited state ($n = 3$).

(c) The energy of the photon is not equal to the excitation energy, so the atom cannot be excited.

Google Forms Microsoft Forms

Quick check 11

→ E-version in OUP Exercise Platform is also available.

1 The energy level of a hydrogen atom in state a is –1.51 eV and its energy level in the ground state is –13.6 eV. Determine whether the following statements are true.

(a) A photon of energy 13 eV can excite the atom from the ground state to state a. (T/F)

(b) An electron of kinetic energy 13 eV can excite the atom from the ground state to state a. (T/F)

(c) A photon of energy 2 eV can ionize the atom in state a. (T/F)

Writing practice

The emission spectrum of hydrogen gas consists of discrete bright lines.



Question 1

Use Bohr's model to explain why the lines are discrete. (2 marks)

Answer

When an atom transits from a higher energy level to a lower one, a photon with energy equal to the energy difference between the two levels is emitted. 1A

Since the energy levels are quantized, the energy, hence the wavelengths, of the photons emitted can be some discrete values only. 1A

Question 2

Why can Rutherford's model not explain the discrete bright lines? (2 marks)

Answer

In Rutherford's model, the electron can revolve around the nucleus with any frequency. 1A

Therefore, the atoms are expected to emit EM waves of a continuous range of frequency. 1A

Practise yourself

The light from a filament lamp passes through some hydrogen gas. The spectrum of the light shows some discrete dark lines.



1 Use Bohr's model to explain why the dark lines are discrete. (2 marks)

2 Explain one limitation of Rutherford's model. (3 marks)

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5 Limitations of Bohr's model

Bohr's model can also explain the spectra of He^+, Li^{2+} and Be^{3+}. They are all ions with only one electron.

Bohr's model can explain the emission and absorption spectra of the hydrogen atom and of a few ions, but it still has some limitations.

1. It fails to explain the spectrum of an atom with more than one electron.
2. It fails to explain the fine structure of the spectral lines, even for hydrogen (many of the spectral lines are in fact not single lines, but very closely spaced combinations of two or more lines).
3. It fails to explain the relative brightness of each spectral line.
4. It does not provide the reason why the angular momentum of the electron is equal to integral multiples of $\frac{h}{2\pi}$.

Although Bohr's model did not provide a complete picture of the atomic structure, it was an important step towards the modern model of the atom, and it played an important role in revealing the quantum nature of matter.

Exercise 2.2

Take $c = 3.00 \times 10^8$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C and $h = 6.63 \times 10^{-34}$ J s.

1 The emission spectra for three gases and the absorption spectrum of a mixture of gases are shown in Figure a.



Fig a

Which gases does the mixture consist of?

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★**2** Which of the following statements is/are correct?

(1) Different elements have their own characteristic emission spectra.
(2) The bright lines in the emission spectrum have the same frequencies as the dark lines in the absorption spectrum of the same element.
(3) If the characteristic spectral lines of an element appear as dark lines in the absorption spectrum of a sample, it can be concluded that the element is absent in the sample.

A (1) only
B (1) and (2) only
C (1) and (3) only
D (2) and (3) only

★**3** In the hydrogen atom, what is the ratio of the radii of the electron's orbits corresponding to quantum numbers $n = 2$ and $n = 4$?

A 1 : 2
B 1 : 3
C 1 : 4
D 1 : 9

★**4** In the hydrogen atom, what is the ratio of the angular momentum of the electron in the first excited state to that in the second excited state?

A 1 : 2
B 2 : 3
C 1 : 4
D 4 : 9

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★**5** Figure b shows some energy levels of an atom (drawn to scale).

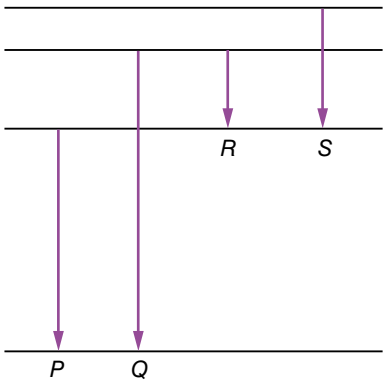


Fig b

Which electron transition will emit electromagnetic waves of the longest wavelength?

A *P*

B *Q*

C *R*

D *S*

★**6** Figure c shows some energy levels of a hydrogen atom (**not** drawn to scale). Photons of frequencies f_P, f_Q, f_R and f_S are emitted due to the electron transitions shown.

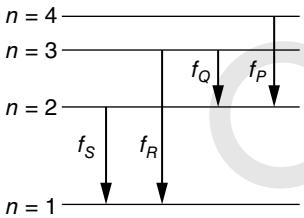


Fig c

Which of the following relationship is/are correct?

(1) $f_S > f_P$

(2) $f_R = f_Q + f_S$

(3) $\frac{1}{f_R} = \frac{1}{f_Q} + \frac{1}{f_S}$

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

★**7** (a) Explain the formation of the emission spectrum of hydrogen.

(b) Explain the formation of the absorption spectrum of hydrogen.

★**8** State one limitation of Rutherford's atomic model and explain how Bohr tackled it.

★**9** (a) Find the total energy possessed by the electron in a hydrogen atom and the radius of its orbit when it is in the 6th excited state. Given: $r_1 = 5.29 \times 10^{-11}$ m

(b) Why is the energy possessed by the electron negative?

★**10** Figure d shows some energy levels of an atom.

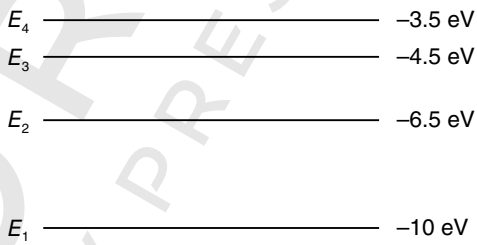


Fig d

(a) In Figure d, draw arrows to represent all electron transitions from higher energy states to state E_2.

(b) Find the frequency and wavelength of the photon emitted in each transition in (a).

(c) Can light of wavelength 150 nm be absorbed by the atom at state E_2? Explain.

★**11** Hydrogen atoms in the ground state are bombarded by electrons each with kinetic energy 12.5 eV. Which energy state(s) can the hydrogen atoms be excited to? Show your calculation.

★**12** Can a photon of energy 3.4 eV ionize a hydrogen atom in

(a) the ground state,

(b) the 1st excited state,

(c) the 2nd excited state.

Show your calculation.

★**13** Find the wavelengths of the four visible spectral lines in the emission spectrum of hydrogen. It is given that these four lines correspond to the electron transitions with final state of $n = 2$ and the wavelength of visible light ranges from 400 nm to 700 nm.

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2.3 Particles or waves?

You learned in Chapter 1 that light exhibits both wave and particle natures. Do particles (e.g. electrons) also exhibit wave nature? What will happen if a beam of electrons is directed onto a double slit (Fig 2.3a)?

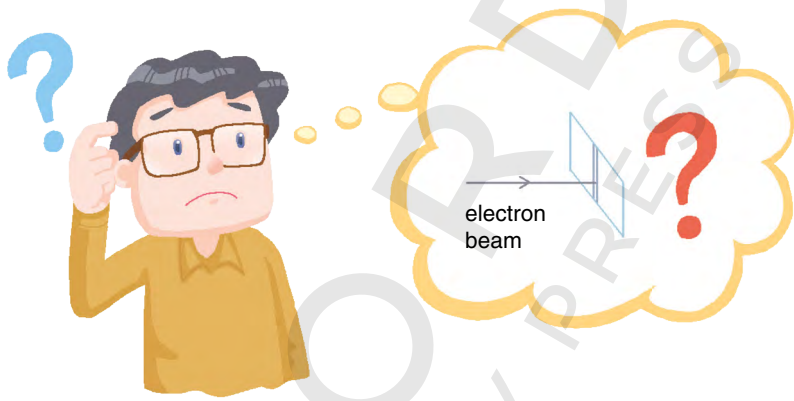


Fig 2.3a *What will happen if a beam of electrons is directed onto a double slit?*

DSE goal

Realize the wave-particle duality of electrons and light.

Relate the wave and particle properties of electrons using $\lambda = h/p$.

Apply $\lambda = h/p$ to solve problems.

DSE exam

12(2)Q2.6, 14(2)Q2.7, 15(2)Q.2(c), 16(2)Q2.5, 17(2)Q2.5, 19(2)Q2.6, 21(2)Q2.6, 23(2)Q2.6, Q.2(a)

1 De Broglie wavelength

In 1917, Einstein assigned to each photon a linear momentum of magnitude given by

$$p = \frac{h}{\lambda}$$

momentum of photon (kg m s^{-1}); Planck constant = 6.63×10^{-34} J s; **wavelength** of light (m)

By $E = \frac{hc}{\lambda}$,

$p = \frac{h}{\lambda} = \frac{E}{c}$

In 1924, *Louis de Broglie* suggested that the wave-particle duality also applied to material particles, such as electrons. Based on symmetry, de Broglie postulated that the equation above could be used to find the wavelength of a moving object with momentum p.

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

wavelength of object (m); **momentum** of object (kg m s^{-1}); Planck constant = 6.63×10^{-34} J s; **speed** of object (m s^{-1}); **mass** of object (kg)

The wavelength calculated is called the **de Broglie wavelength**. The larger the momentum (larger mass or higher speed), the shorter the wavelength.

Louis de Broglie 德布羅意　　de Broglie wavelength 德布羅意波長

Know more

Louis de Broglie (1892–1987)



De Broglie was awarded the Nobel Prize in Physics in 1929 for his discovery of the wave nature of electrons.

Example 9 Momentum of a photon

Find the momentum of a photon of

(a) red light of wavelength 630 nm,

(b) blue light of wavelength 450 nm.

Solution

(a) $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{630 \times 10^{-9}} = 1.05 \times 10^{-27}$ kg m s^{-1}

(b) $p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{450 \times 10^{-9}} = 1.47 \times 10^{-27}$ kg m s^{-1}

Example 10 Wavelengths of moving objects

(a) Find the de Broglie wavelengths of

(i) a tennis ball of mass 50 g moving at 60 m s^{-1},



(ii) an electron of mass 9.11×10^{-31} kg moving at 5×10^{6} m s^{-1}.

(b) How will the de Broglie wavelength of an object change if its speed increases?

Solution

(a) (i) $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(0.05)(60)} = 2.21 \times 10^{-34}$ m

The wavelength of objects in daily life is so short that wave phenomena like diffraction and interference cannot be observed. ▶

(ii) $\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{(9.11 \times 10^{-31})(5 \times 10^{6})} = 1.46 \times 10^{-10}$ m

The wavelength of a fast-moving electron is comparable to the atomic spacing (separation between nuclei of nearby atoms). ▶

(b) According to $\lambda = \frac{h}{mv}$, λ will decrease if v increases.

Think about...

What would the speed of a tennis ball be if its de Broglie wavelength is 1 m? How long would the ball take to move 1 cm? Can we observe the wave nature of the tennis ball?

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Quick check 12

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

Take $h = 6.63 \times 10^{-34}$ J s.

1 Complete the following table about the momentum of a photon.

	λ / nm	p / kg m s^{-1}
Violet light	400	
Yellow light		1.14×10^{-27}

2 Complete the following table about the de Broglie wavelength.

	λ / m	m / kg	v / m s^{-1}
Bird		0.03	10
Man	2.21×10^{-36}	60	

Example 11 Wavelengths of electrons with known KE

(a) A particle of mass m moves with momentum p and kinetic energy K.

(i) Express K in terms of m and p.

(ii) Express its de Broglie wavelengths in terms of m and K.

(b) An electron of mass 9.11×10^{-31} kg has kinetic energy 10 keV.

(i) Find its de Broglie wavelength.

(ii) What will be its new de Broglie wavelength if its kinetic energy is doubled?

Solution

(a) (i) $K = \frac{1}{2}mv^2 = \frac{1}{2m}(m^2v^2) = \frac{p^2}{2m}$

(ii) From (i), $p = \sqrt{2mK}$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

(b) (i) $\lambda = \frac{h}{\sqrt{2mK}}$

$$= \frac{6.63 \times 10^{-34}}{\sqrt{2(9.11 \times 10^{-31})(10 \times 1000)(1.6 \times 10^{-19})}}$$

$$= 1.228 \times 10^{-11} \text{ m}$$

$$\approx 1.23 \times 10^{-11} \text{ m}$$

(ii) Since $\lambda = \frac{h}{\sqrt{2mK}} \propto \frac{1}{\sqrt{K}}$,

$$\frac{\lambda_2}{\lambda_1} = \frac{\sqrt{K_1}}{\sqrt{K_2}}$$

$$\lambda_2 = \frac{\sqrt{K_1}}{\sqrt{K_2}} \times \lambda_1 = \frac{\sqrt{1}}{\sqrt{2}} \times 1.228 \times 10^{-11} = 8.68 \times 10^{-12} \text{ m}$$

▶ Exercise 2.3 Q7(b) (p.81)

Example 12 Wavelength of an electron in a hydrogen atom

The electron in a hydrogen atom is in an orbit with a quantum number $n = 3$. Find its de Broglie wavelength. The radius of the innermost electron orbit is 5.29×10^{-11} m.

Solution

By $m_e v_n r_n = \frac{nh}{2\pi}$,

$$2\pi r_n = \frac{nh}{m_e v_n}$$

$$2\pi(n^2 r_1) = n\lambda_n \qquad \lambda_1 = 2\pi(1)r_1 = 2\pi r_1$$

$$\lambda_n = 2\pi n r_1 \qquad \therefore \lambda_n = n(2\pi r_1) = n\lambda_1$$

$$\therefore \lambda_3 = 2\pi(3)(5.29 \times 10^{-11}) = 9.97 \times 10^{-10} \text{ m}$$

▶ Exercise 2.3 Q7(c) (p.81)

→ This simulation shows what orbits are allowed in Bohr's model. It is available on OUP web in PC version (.exe) only.

Simulation Video

→ This video shows the stationary waves formed in a closed circular-wire loop.

If n is not an integer, the wave will undergo destructive interference with itself and will die out quickly.

With his hypothesis, de Broglie was able to explain why the electron in a hydrogen atom could only have quantized angular momentum $\frac{nh}{2\pi}$. He proposed that the electron in the hydrogen atom behaves as a stationary wave. The length of its orbit must be an integral multiple of its de Broglie wavelength (Fig 2.3b):

$$n\lambda_n = 2\pi r_n \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (6)$$

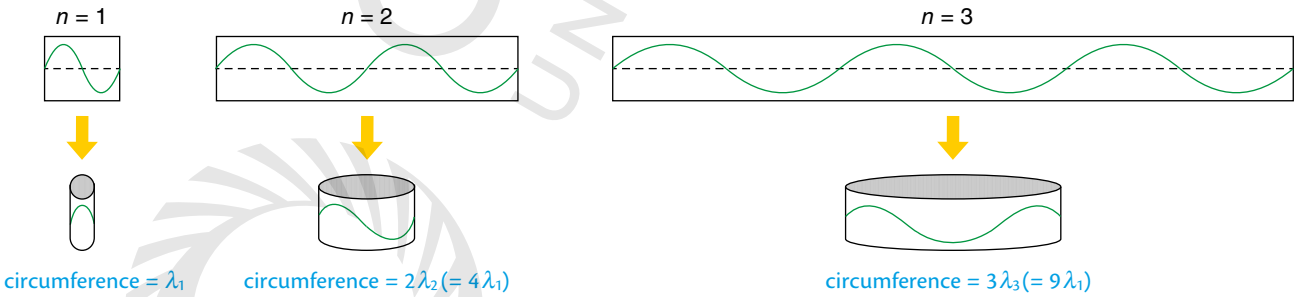


Fig 2.3b *Length of orbit and de Broglie wavelength of an electron.*

Substitute $\lambda_n = \frac{h}{m_e v_n}$ into (6),

$$\frac{nh}{m_e v_n} = 2\pi r_n$$

$$m_e v_n r_n = \frac{nh}{2\pi}$$

Therefore, the angular momentum of the electron is quantized.

DSE goal

Describe evidence of electrons exhibiting both wave and particle properties.

DSE exam

16(2)Q2.2

2 Evidence of the wave nature of matter

The wave nature of matter can be verified by experiments.

1 When a beam of electrons is directed at a thin metal foil (Fig 2.3c), a diffraction pattern consisting of bright and dark rings is formed (Fig 2.3d).

Watch a video of an experiment in which a beam of electrons is directed onto a thin piece of graphite.

Video

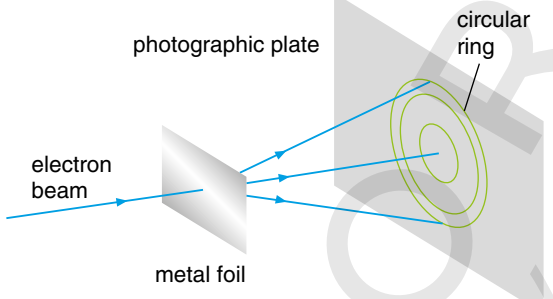


Fig 2.3c *A beam of electrons directed onto a metal foil.*

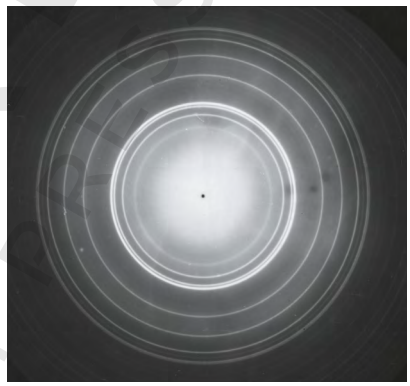


Fig 2.3d *Electron diffraction pattern.*

According to the result of Example 10 on p.77, the de Broglie wavelength of fast-moving electrons is comparable to the atomic spacing, which is a necessary condition for diffraction to occur.

Know more

Wave nature of electrons

Experiments were perfomed in 1927 to verify the wave nature of electrons.

▸ p.94

2 When a beam of electrons is directed onto a double slit with a sufficiently small slit width, an interference pattern is obtained (Fig 2.3e). Similar experiment results have been obtained for other particles (e.g. neutrons, protons).

→ The following video shows how the interference pattern is built up.



Fig 2.3e *An interference pattern formed by electrons passing through a double slit.*

These experiments give strong support to the wave nature of electrons and other particles.

Quick check 13

→ E-version in OUP Exercise Platform is also available.

Google Forms

Microsoft Forms

1 *True or false:*

(a) Interference of light shows the wave nature of light. (T/F)

(b) Photoelectric effect shows the particle nature of light. (T/F)

(c) Electron diffraction by a thin metal foil shows the particle nature of electrons. (T/F)

Exercise 2.3

Take $c = 3.00 \times 10^8$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $h = 6.63 \times 10^{-34}$ J s.

★**1** Two particles X and Y are travelling at the same speed. The de Broglie wavelength of particle X is λ. If the mass of particle X is half that of particle Y, what is the de Broglie wavelength of particle Y?

A $\frac{\lambda}{4}$

B $\frac{\lambda}{2}$

C 2λ

D 4λ

★**2** The momentum of a photon is p (in kg m s^{-1}). How much energy does the photon possess?

A $\frac{h}{p}$ (in J)

B $\frac{hc}{p}$ (in J)

C $\frac{hc}{p}$ (in eV)

D pc (in J)

★**3** Which of the following are evidence of electrons exhibiting wave properties?

(1) Bright and dark rings are formed when a beam of electrons passes through a metal foil.

(2) Bright and dark fringes are formed when a beam of electrons passes through a double slit.

(3) Photoelectrons are emitted when light falls on a material.

A (1) and (2) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**4** Which of the following photon has the largest momentum?

A A photon of radio waves

B A photon of microwaves

C A photon of infra-red radiation

D A photon of gamma rays

★**5** Find the de Broglie wavelengths of the following objects. Can the wave nature of the objects be observed? Explain.

(a) A kitten of 300 g walking at 10 cm s^{-1}



(b) A car of 1500 kg moving at 60 km h^{-1}



★**6** An electron is accelerated from rest until its de Broglie wavelength becomes 1.84×10^{-10} m.

(a) During the acceleration, how does the de Broglie wavelength of the electron change?

(b) How fast does the electron travel finally?

★**7** Find the de Broglie wavelengths of the following particles.

(a) An α particle of mass 6.64×10^{-27} kg moving at 15 000 km s^{-1}

(b) An electron with kinetic energy 6×10^4 eV

(c) The electron in a hydrogen atom in the 1st excited state (given: radius of the innermost electron orbit = 5.29×10^{-11} m)

★**8** (a) Find the momentum of an X-ray photon of frequency 8.6×10^{17} Hz.

(b) The momentum of a photon is 2.18×10^{-27} kg m s^{-1}. How much energy (in J) does the photon carry?

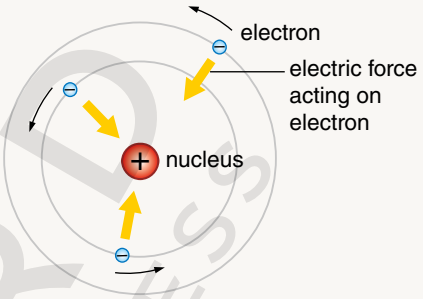
(c) A photon carries 3 eV of energy. Find its momentum.

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Key concepts 2

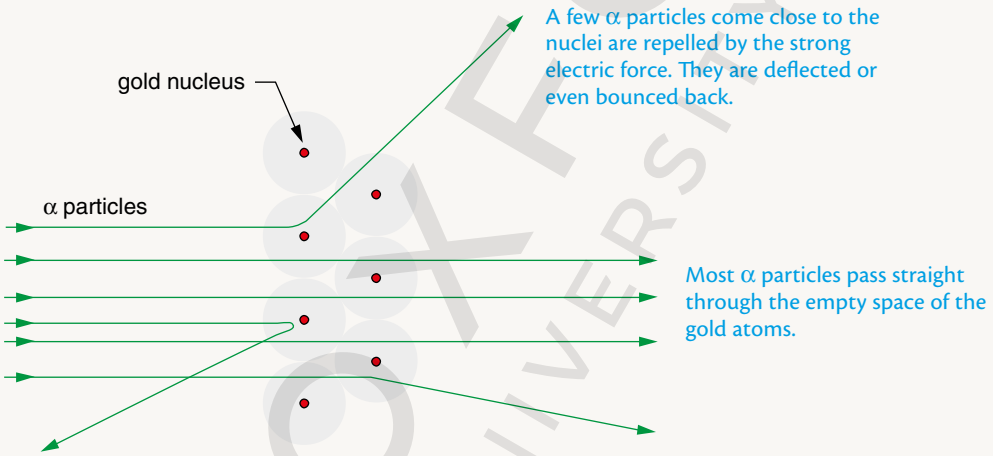
1 Rutherford's model

- All the positive charge and most of the mass are concentrated at the nucleus. The rest of the atom is largely empty space.
- Electrons orbit the nucleus. The centripetal force is provided by the electric force acting on it by the nucleus.



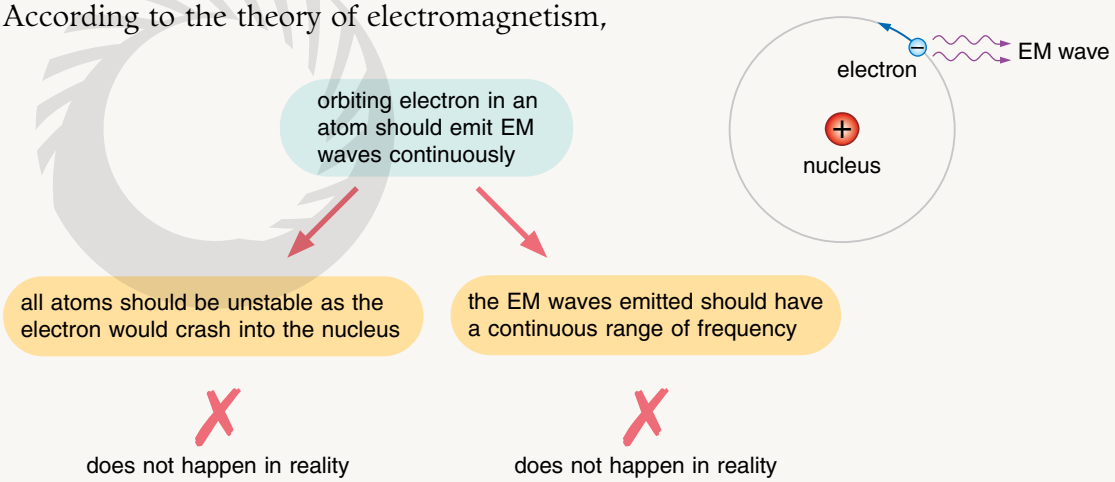
2 Explaining the results of Geiger-Marsden experiment

In the experiment, a thin gold foil was bombarded with α particles.



3 Limitations of Rutherford's model

According to the theory of electromagnetism,



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4 Bohr's postulates of the hydrogen atom

Classical aspect

1 The electron is a particle moving around the nucleus in a circular orbit. The centripetal force is provided by the electric force.

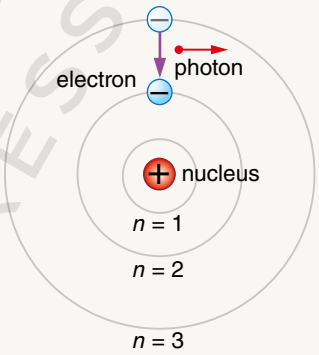
Quantum aspects

2 The electron can only occupy certain discrete orbits.

3 When the electron stays in an orbit, it does not radiate and its total energy remains unchanged.

4 The electron can transit from one orbit to another by gaining or losing energy. This can be done by absorbing or emitting a photon.

5 The angular momentum of the electron is quantized in integral multiples of $\frac{h}{2\pi}$.



5 Bohr's model of the hydrogen atom

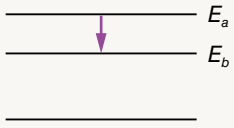
Discrete orbits and energy levels

When the electron is in the nth orbit,

- quantum number = n
- radius of orbit $r_n = n^2 r_1$
- angular momentum $m_e v_n r_n = \frac{nh}{2\pi}$
- energy $E_n = -\frac{13.6}{n^2}$ eV

Emission

Energy is emitted as a photon when an electron transits from a higher energy level E_a to a lower energy level E_b.



$$hf = \frac{hc}{\lambda} = E_a - E_b$$

$$\frac{1}{\lambda} = \frac{13.6 \text{ eV}}{hc}\left(\frac{1}{b^2} - \frac{1}{a^2}\right)$$

Absorption

An electron can jump from a lower energy level E_b to a higher energy level E_a by

- absorbing a photon with energy exactly equal to $E_a - E_b$,
- colliding with a particle with kinetic energy higher than $E_a - E_b$.

Ionization

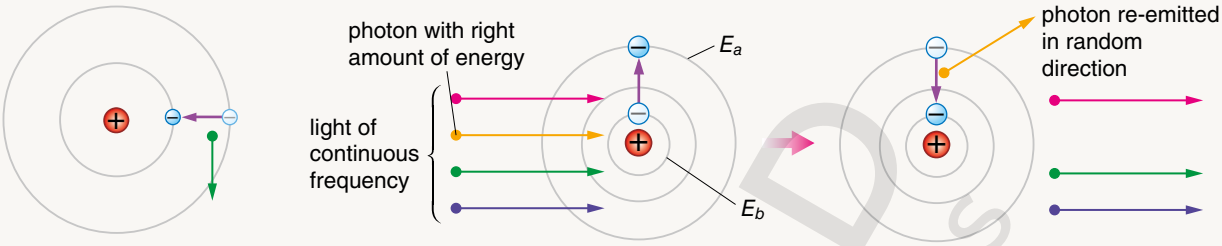
When the electron is at energy level E_n,

- minimum energy $E_{\min}$ to ionize the atom is given by $E_\infty - E_n$;
- the atom can be ionized by absorbing an amount of energy higher than $E_{\min}$.

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6

Emission spectrum and absorption spectrum



Emission spectrum: Photons of certain energies (frequencies) are emitted by electrons dropping to lower energy levels, producing discrete bright lines

Absorption spectrum: When light of continuous frequency is sent through a gas, photons of certain energies (frequencies) may be absorbed, so as to raise the electron from a lower energy level to a higher energy level. This produces dark lines in a continuous spectrum.



The frequencies of spectral lines are determined by the discrete energy levels of the atoms and are therefore the same for the same element.

7

Wave nature of matter

De Broglie wavelength:

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

De Broglie proposed that the electron in a hydrogen atom behaved as a wave and formed stationary wave along its orbit, which gives:

$$n\lambda = 2\pi r_n$$

$$\Rightarrow m_e v_n r_n = \frac{nh}{2\pi} \quad \text{(Bohr's postulate)}$$

Key terms

1	absorption spectrum 吸收光譜	p.52
2	angular momentum 角動量	p.57
3	Bohr's model 玻爾模型	p.56
4	de Broglie wavelength 德布羅意波長	p.76
5	electron transition 電子躍遷	p.63
6	emission spectrum 發射光譜	p.51
7	energy level 能級	p.53
8	excitation energy 激發能	p.63
9	excited state 受激態	p.61
10	ground state 基態	p.61
11	ionization energy 電離能	p.66
12	line spectrum 線狀光譜	p.51
13	quantum number 量子數	p.56

Revision 2

Take $c = 3.00 \times 10^8$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, $h = 6.63 \times 10^{-34}$ J s.

Multiple-choice questions

★1 2.1 Which of the following can be concluded from the results of the Geiger-Marsden experiment?

(1) The mass in an atom is not evenly distributed.

(2) An atom is not neutral.

(3) The electrons in an atom orbit the nucleus.

A (1) only

B (3) only

C (1) and (2) only

D (1), (2) and (3)

★2 2.2 The figure shows some energy levels of an atom (drawn to scale) (Fig a).

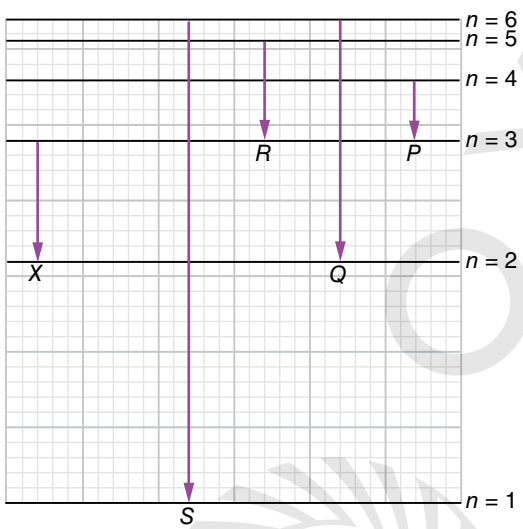


Fig a

If transition X results in the emission of a photon of wavelength 750 nm, which transition would result in the emission of a photon of wavelength 375 nm?

A P

B Q

C R

D S

★3 2.2 Figure b shows some energy levels of atom P.

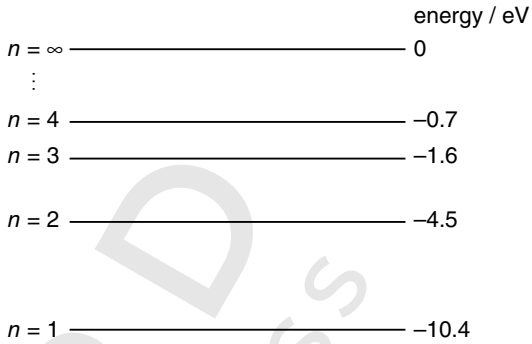


Fig b

If photons having energy of 8.9 eV are incident on a large number of atom P at the first excited state ($n = 2$), how many spectral lines in the visible region can be obtained subsequently? Given: the energy of a visible light photon ranges from 1.7 eV to 3.2 eV.

A 0

B 1

C 2

D 6

☆ HKDSE 2018 Paper 2 Q2.1

★4 2.1 An α particle approaches a gold nucleus. Its path is shown in Figure c.

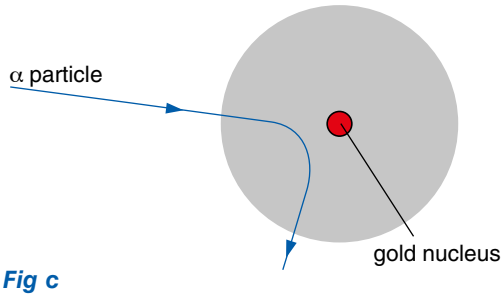


Fig c

Which of the following statements is/are correct?

(1) When the α particle gets closer to the gold nucleus, the force that it experiencs increases.

(2) When the α particle moves along its path, its KE is constant.

(3) The shortest distance between the α particle and the nucleus gives an estimation of the upper limit of the radius of the nucleus.

A (1) only

B (3) only

C (1) and (2) only

D (1) and (3) only

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★5 2.2 Which of the following statements about the electron in a hydrogen atom is/are correct? It is given that the energy of the electron is –13.6 eV when it is in the ground state.

(1) When the electron is in the ground state, it can escape completely from the nucleus by absorbing a photon of energy 14 eV.

(2) When the electron is in the ground state, it can be excited to the first excited state by absorbing a photon of energy 11 eV.

(3) When the electron drops from an excited state to the ground state, it must emit a photon of energy 13.6 eV.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

★6 2.3 Which of the following could be explained by the wave nature of electrons?

(1) A pattern consisting of bright and dark rings is observed when an electron beam is directed onto a thin metal foil.

(2) The electron in a hydrogen atom can only occupy certain discrete orbits with quantized angular momentum.

(3) When an electron is orbiting in an atom, the electric force between the electron and the nucleus provides the centripetal force.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

★7 2.2 A photon of violet light is emitted when a hydrogen atom drops from a higher energy state with quantum number n to the first excited state. If the wavelength of the photon is estimated to be 4.11×10^{-7} m, what is n?

A 3 B 4

C 5 D 6

★8 2.2 What is the maximum wavelength of the light that can ionize a hydrogen atom in its second excited state? The ground state energy of the atom is –13.6 eV.

A 103 nm B 122 nm

C 165 nm D 823 nm

★9 2.3 Two objects have the same de Broglie wavelength. Which of the following quantities must be the same for the two objects?

(1) Momentum

(2) Velocity

(3) Kinetic energy

A (1) only B (2) only

C (1) and (3) only D (1), (2) and (3)

★10 2.2 Which of the following particles may collide inelastically with a hydrogen atom in the first excited state?

(1) An electron of KE 1.6 eV

(2) A proton of KE 2 eV

(3) A neutron of KE 3.5 eV

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★11 2.2 When charged particles, such as electrons, emitted by the Sun collide with gas particles in the Earth's atmosphere, the gas particles may be excited. When these excited particles drop to lower energy levels, photons are emitted, forming aurora (Fig d).



Fig d

Suppose the gas particles are in the ground state. Which of the following is/are the possible speed(s) of the electron that can produce red light of wavelength 630 nm?

(1) 1000 km s^{-1}

(2) 2000 km s^{-1}

(3) 3000 km s^{-1}

A (3) only

B (2) and (3) only

C (1), (2) and (3)

D None of the above

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12 2.2 The electron in a hydrogen atom transits from the first excited state to the second excited state. How do its total energy E_T and KE change?

	E_T	KE
A	increase	increase
B	increase	decrease
C	unchanged	increase
D	unchanged	decrease

13 2.2 Figure e shows three energy levels of a certain atom.

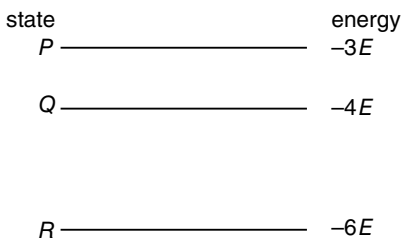


Fig e

When the atom drops from state P to Q, a photon of wavelength λ is emitted. What is the wavelength of the photon emitted when the atom drops from state Q to R?

A $\frac{1}{2}\lambda$

B $\frac{2}{3}\lambda$

C $\frac{3}{2}\lambda$

D 2λ

14 2.3 Two particles X and Y have the same mass and the kinetic energy of particle X is E. If the de Broglie wavelength of particle X is double that of particle Y, what is the kinetic energy of particle Y?

A $\frac{E}{4}$

B $\frac{E}{2}$

C $2E$

D $4E$

Refer p.78

15 Misc A light ray of power 20 W is shone on a mirror perpendicularly (Fig f).

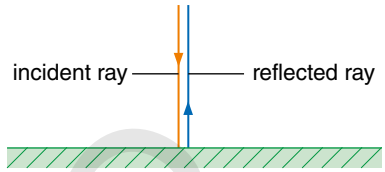


Fig f

Assuming all photons are reflected, what is the magnitude of the average force exerted on the mirror by the photons?

A 0

B 6.67×10^{-8} N

C 1.33×10^{-7} N

D Cannot be determined as the frequency of light is unknown

Refer p.17, 76, Book 2 p.300

16 2.1 **HKDSE 2016 Paper 2 Q2.1**

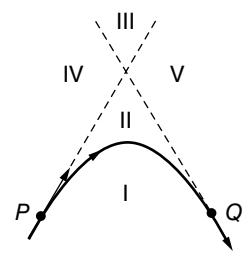


Fig g

In Figure g, the solid line is the trajectory of an α particle scattered by a gold nucleus (not shown in figure). The dotted lines are tangents to the trajectory at points P and Q. The two dotted lines together with the trajectory divide the plane into five regions (I–V). In which region(s) can the gold nucleus be situated?

A I B II

C III (34%) D IV or V

17 2.2 **HKDSE 2019 Paper 2 Q2.1**

There are dark lines in the Sun's spectrum because lights at certain wavelengths emitted by the Sun are

A completely absorbed by the Sun's atmosphere.

B completely absorbed by the Earth's atmosphere.

C partly absorbed by the Sun's atmosphere. (23%)

D partly absorbed by the Earth's atmosphere.

18 **HKDSE 2020 Paper 2 Q2.1**
2.1

Which of the following statements about Rutherford's scattering experiment is/are correct?

(1) Thin gold foil was used so that each α particle would most likely be scattered by one gold nucleus only.

(2) If β particles were directed towards the gold foil instead of α particles, they could be deflected by both the gold nuclei and the electrons in the foil.

(3) Gold foil was used because gold contains free electrons.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only (29%)

19 **HKDSE 2020 Paper 2 Q2.3**
2.2

The figure below shows the emission lines of the hydrogen spectrum associated with electron transitions from excited energy levels to the ground state (n = 1). It is known that the energy level of the ground state is E_1 = –13.6 eV.

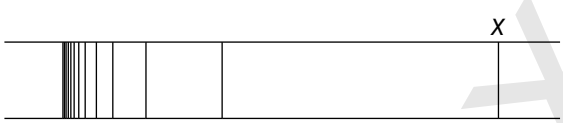


Fig h

Which statement is correct?

A Spectral line X has the highest frequency.

B The shortest wavelength of the lines in this spectral series is about 90 nm. (27%)

C These spectral lines are in the infra-red region.

D Some of these lines also appear in other spectral series of the hydrogen spectrum.

20 **HKDSE 2020 Paper 2 Q2.4**
2.2

Which statements about the Bohr atom model are correct?

(1) It can explain why α particles can be rebounded by a thin gold foil.

(2) It can give the atomic spectra of a singly ionized helium atom (He^+).

(3) A postulate of this model is that the angular momentum of the electron in a hydrogen atom is quantized.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3) (23%)

Conventional questions

★**21** Misc

The result of the Geiger-Marsden experiment gives evidence to the existence of a nucleus in a gold atom (Fig i).

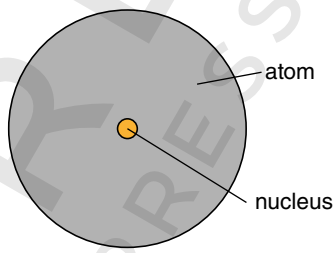


Fig i

(a) Which type of charge does the nucleus carry? (1 mark)

(b) State the observations in the Geiger-Marsden experiment which lead to the conclusion in (a). Explain briefly. (3 marks)

(c) (i) How did Rutherford describe the parts in an atom other than the nucleus? (2 marks)

(ii) State two limitations of Rutherford's atomic model. (2 marks)

(d) State one similarity and one difference between Rutherford's model and Bohr's model. (2 marks)

★**22** 2.2

The light emitted by the Sun passes through the Sun's atmosphere and produces an absorption spectrum (Fig j).

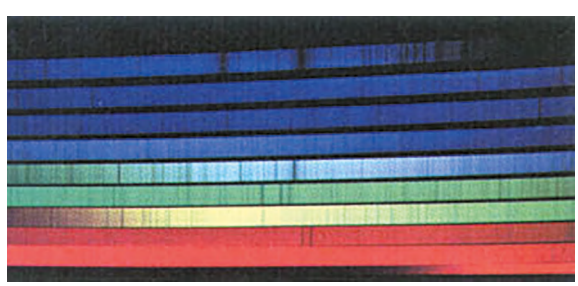


Fig j

(a) Explain how the dark lines in the absorption spectrum are formed. (3 marks)

(b) How can the elements in the Sun's atmosphere be identified by studying these dark lines? (2 marks)

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(c) The dark line of frequency 6.89×10^{14} Hz corresponds to the transition that hydrogen atoms are excited from state $n = 2$ to state $n = a$. Find a. (2 marks)

★**23** Misc Figure k shows some energy levels of an atom of a gas.

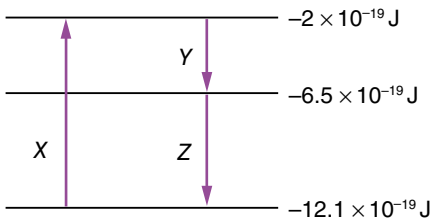


Fig k

(a) A photon is absorbed to trigger electron transition X.

(i) Find its frequency. (2 marks)

(ii) Find the magnitude of its momentum. (2 marks)

(b) After transition X takes place, the electron drops back to the original energy level by transitions Y and Z sequentially. Determine whether the transitions emit visible light and explain your answer. Given: the wavelength of visible light ranges from 400 nm to 700 nm. (3 marks)

(c) When light of continuous frequency passes through this gas, a spectrum of the gas is observed from the other side of the gas.

(i) State whether it is an emission spectrum or an absorption spectrum. (1 mark)

(ii) Describe what the spectrum looks like. (2 marks)

★**24** 2.3 (a) An electron is accelerated so that it possesses KE of 50 000 eV.

(i) Find the magnitude of the momentum of the electron. (2 marks)

(ii) Hence find the de Broglie wavelength of the electron. (2 marks)

(b) Describe an observation that shows the wave nature of electrons. (1 mark)

(c) Estimate the shortest de Broglie wavelength of the electron in a hydrogen atom. The radius of the innermost orbit of the electron is 5.29×10^{-11} m. (2 marks)

★**25** 2.2 Figure l shows the simplified structure of a fluorescent lamp. The inner wall of the tube is coated with phosphor powder. Inside the tube, there are an inert gas and a small amount of mercury.

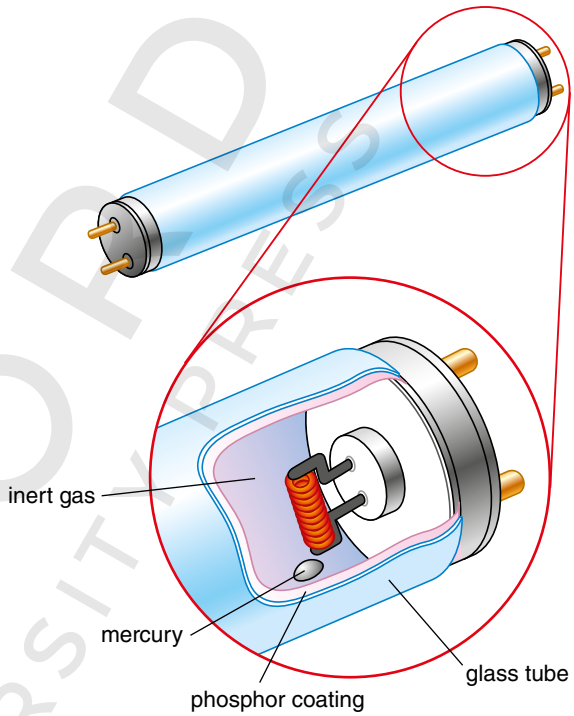


Fig l

When the lamp is switched on, the gaseous mercury atoms are excited. Then the excited atoms return to the ground state and emit ultra-violet radiation of a wavelength of about 256 nm.

(a) Estimate the energy of a photon of the ultra-violet radiation emitted in eV. (2 marks)

(b) How could the phosphor coating convert the ultra-violet radiation into visible light? Explain briefly with the energy level diagram of phosphor in Figure m. Given: the energy of a visible light photon ranges from 1.7 eV to 3.2 eV. (4 marks)

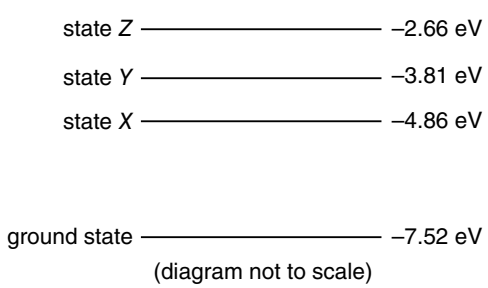


Fig m

★**26** 2.2 Figure n shows the energy levels of a hydrogen atom (**not** to scale).

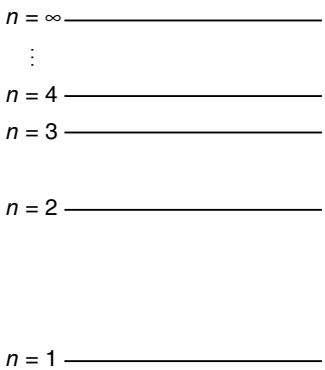


Fig n

When the hydrogen atoms drop from higher energy levels to lower ones, radiation of discrete frequencies are emitted.

(a) Why are the frequencies of the radiation emitted discrete? (2 marks)

(b) (i) In Figure n, draw two arrows to indicate the electron transitions, from excited states to the ground state, that emit photons of the highest and lowest frequencies. (2 marks)

(ii) Hence, find the two frequencies. (3 marks)

(c) What does the state $n = \infty$ represent? (1 mark)

★**27** 2.2 Figure o shows some energy levels of an electron in a certain atom.

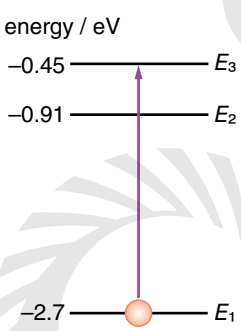


Fig o

The electron is excited from E_1 to E_3 after absorbing a photon.

(a) Find the wavelength of the photon. (2 marks)

(b) Red light is emitted when the electron transits to a lower energy level.

(i) What is the corresponding transition? (1 mark)

(ii) Find the wavelength of the red light. (1 mark)

(c) Can the electron at energy level E_1 absorb a photon of frequency

(i) 5×10^{14} Hz, (2 marks)

(ii) 7×10^{14} Hz? (2 marks)

★**28** Misc In a hydrogen atom, the electron is moving in the orbit with quantum number $n = 3$. The radius of the innermost orbit is 5.29×10^{-11} m.

(a) Find the de Broglie wavelength of the electron. (2 marks)

(b) The electron then drops to the innermost orbit by emitting a photon. Find the wavelength of the photon emitted. (2 marks)

(c) How will the results in (a) and (b) change if the electron starts from an orbit with a larger quantum number n? Explain your answer. (3 marks)

★★**29** 2.2 Part of the emission spectrum of hydrogen is sketched in Figure p. X is the group of lines which are produced when the hydrogen atoms drop from higher energy states to the first excited state.

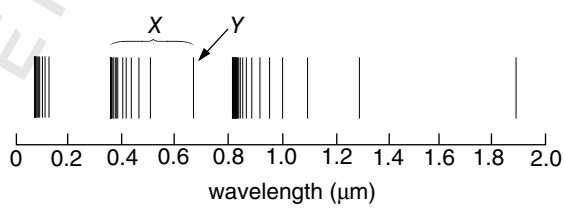


Fig p

(a) (i) State the electron transition which produces spectral line Y. (1 mark)

(ii) Find the wavelength of spectral line Y. (2 marks)

(iii) Which region of the electromagnetic spectrum does line Y belong to? (1 mark)

(iv) Find the shortest wavelength of the lines in X. (1 mark)

(b) Explain briefly how the discovery of the line spectrum of the hydrogen supports the theory of discrete energy levels in atoms. (3 marks)

▶ Refer p.70

See p.92 for the Exam Report.

30 **HKDSE 2015 Paper 2 Q.2**

Misc

The energy levels E_n of an electron in a hydrogen atom from the Bohr model can take the form below:

$$E_n = \frac{-13.6}{n^2} \text{ eV where } n = 1, 2, 3, \ldots$$

(a) Bohr's idea was sometimes criticized by some physicists as semi-classical and semi-quantum. Point out **ONE classical aspect** in the Bohr model. (1 mark)

(b) From the energy point of view, state the physical meaning of a hydrogen atom being in its **ground state**. (1 mark)

(c) If the minimum energy required to ionize a hydrogen atom in the ground state is E, express the minimum momentum p of a photon for ionizing such a hydrogen atom in terms of E and another physical constant. (2 marks)

(d) Hydrogen atoms in the ground state are bombarded by electrons each with kinetic energy 12.9 eV.

(i) Show that these hydrogen atoms can be excited at most to the third excited state (i.e. $n = 4$). (2 marks)

(ii) For a hydrogen atom excited to the third excited state ($n = 4$), what is the de Broglie wavelength of the orbiting electron in the atom? Given: the orbital radius r_n of the electron in a hydrogen atom from the Bohr model is equal to $0.053\ n^2$ (unit: nm), where $n = 1, 2, 3, \ldots$ (2 marks)

(iii) Draw arrow(s) in Figure q to illustrate all possible transitions leading to emission of photons by these excited hydrogen atoms. (2 marks)

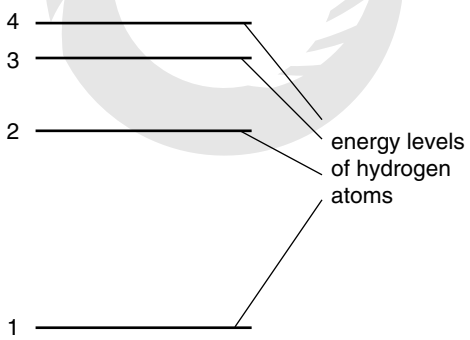


Fig q

See p.92 for the Exam Report.

31 **HKDSE 2019 Paper 2 Q.2**

Misc

(a) In Thomson's 'plum-pudding' model of atoms, an atom consists of a lump of positive material embedded with negatively-charged electrons distributed throughout.

(i) In order to test this atomic model, an experiment was performed such that a beam of α particles was shot at a gold foil and the deflections of the α particles were measured. State the result(s) of this scattering experiment. (2 marks)

(ii) Thomson's atomic model cannot account for the results of the scattering experiment in (a)(i). Why? (1 mark)

(b) The diagram below represents some energy levels of a hydrogen atom (Fig r). The ground state energy E_0 of hydrogen atom is –13.6 eV.

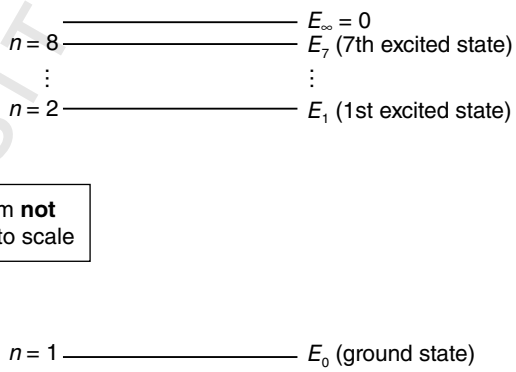


Diagram **not** drawn to scale

Fig r

(i) All energy levels of a hydrogen atom take negative values except E_∞. State the physical significance of energy levels having 'negative values' and the implication of an electron being at E_∞. (2 marks)

(ii) What is the wavelength of electromagnetic wave emitted from a hydrogen atom which undergoes a transition from its 7th excited state ($n = 8$) to the 1st excited state ($n = 2$)? (3 marks)

(iii) Find the minimum energy required to ionize a hydrogen atom from its 3rd excited state (not shown). (2 marks)

32 **HKDSE 2021 Paper 2 Q.2**
Misc

(a) Rutherford's planetary model of the atom failed to account for the stability of atoms. Why? (1 mark)

(b) The emission spectrum of hydrogen atoms only has four visible spectral lines (*A* to *D*) as shown in Figure s.

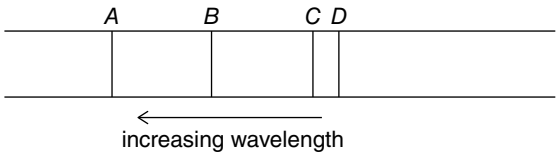


Fig s

All these lines belong to a series that corresponds to transitions to the first excited state ($n = 2$). In this series there are no spectral lines beyond *A*. The wavelengths λ (in nm) of all the spectral lines in the series are given empirically by the formula below.

$$\lambda = 364.6\left(\frac{n^2}{n^2 - 2^2}\right) \text{ where } n = 3, 4, 5, \ldots$$

(i) Which spectral line (*A*, *B*, *C* or *D*) comes from the electron transition between energy levels $n = 5$ and $n = 2$? (1 mark)

(ii) Find the wavelength of the spectral line in (b)(i) and state the colour of this line. (2 marks)

(iii) The remaining numerous invisible spectral lines in the series beyond line *D* are getting closer and closer until they finally converge to a limit of 364.6 nm. Suppose a photon of wavelength shorter than 364.6 nm collides with a hydrogen atom in the first excited state ($n = 2$). State what would happen to **the incident photon**, **the hydrogen atom** and **its orbital electron**. (3 marks)

(iv) Initially a group of hydrogen atoms are in the third excited state ($n = 4$). Illustrate with the aid of an energy level diagram **ALL** possible electron transitions that would produce **emission lines**. Mark a letter 'V' against the transition(s) that give(s) rise to visible spectral lines. (3 marks)

Q30 Exam Report: Part (a) was well answered. Most candidates understood the physical meaning of ground state. In (c), some of the candidates managed to find the relation correctly but just wrote $E = pc$ instead of expressing p explicitly in terms of E and c. (d)(i) was in general well answered. Quite a lot of candidates showed that the energy was large enough to excite the hydrogen atoms to the third excited state but not why there was not enough energy to excite it to the fourth excited state. Some candidates arrived at negative ΔE by using $E_1 - E_4$. In (d)(ii), few candidates were able to apply $n\lambda = 2\pi r$ to find the wavelength. Some assumed that the KE of the electron equals to its PE at E_4 but actually KE = –PE. Not many indicated all the transitions correctly in (d)(iii). Most of them only drew 4 to 5 lines and some with incorrect arrows.

Q31 Exam Report: In (a)(i), it seemed that most candidates knew the results of Rutherford's scattering experiment, however, some failed to provide a concise description regarding the degree of deflection and the amount of alpha particles being deflected. Part (a)(ii) was poorly answered. More than half of the candidates held the belief that the majority of the alpha particles would be rebounded backwards if the Thomson's atomic model were true. In (b)(i), weaker candidates did not know the physical significance of energy levels being negative. Some tried to relate it with the force of attraction between electrons and the nucleus. Candidates' performance in (b)(ii) was satisfactory. Most knew how to find the energy difference corresponding to the electron transition though a few failed to obtain the correct wavelength. In (b)(iii), some candidates just took the energy of the 3rd excited state instead of the difference between this value and E_∞ as the energy for ionizing an atom at that particular energy level.

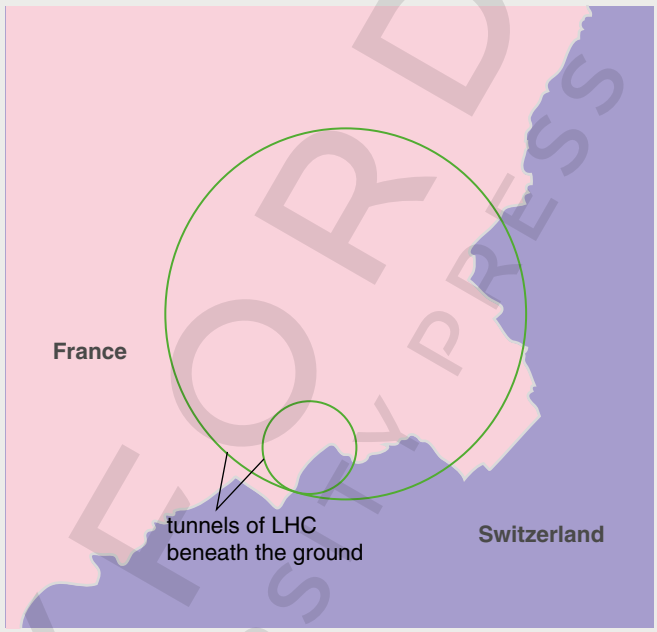
Q32 Exam Report: In (a), most candidates realised that the atom would collapse according to Rutherford's model but some did not give a reason. A few candidates had no idea of the circular motion of the electron and held that it would hit the nucleus due to electrostatic attraction. Candidates did poorly in (b)(i) and the responses were quite random. (b)(ii) was well answered. Most managed to find the wavelength but quite a number of them failed to state the correct colour. Candidates' performance in (b)(iii) was fair. Some did not know the quantum nature of photons and stated that the photon lost some energy to the atom and left. In fact an atom would absorb the whole photon if the photon energy was larger than the ionization energy. Some of them perhaps confused this with excitation in which only a certain amount of energy could be absorbed and concluded that the photon would not be absorbed while the atom was not affected. In (b)(iv), not many were able to mark the two electron transitions for the visible lines correctly. Some candidates omitted those transitions to $n = 1$. Quite a lot of them did not draw the energy level diagram in a conventional manner, i.e. the transitions were not drawn in groups or the diagram was drawn horizontally.

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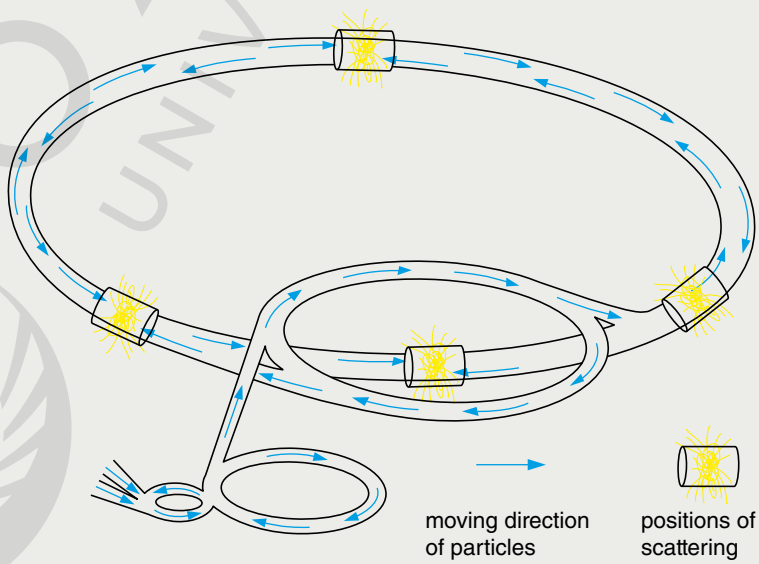
Know more

Particle accelerators and new particles (p.48)

Some modern scattering experiments require particles colliding at very high speeds. This is done inside a particle accelerator. The largest particle accelerator in the world is the Large Hadron Collider (LHC) in Europe. One of its aims is to test the existence of the Higgs boson, which is often referred to as 'the God particle' in the media.



LHC lies in a ring-shaped tunnel 27 km in circumference, as deep as 175 m beneath the ground at the border between France and Switzerland.



The structure of LHC.

Using these advanced particle accelerators, scientists have discovered many new particles, including the quarks, which are the constituting particles of protons and neutrons. These discoveries, which showed that protons and neutrons are not the fundamental units of matter, are crucial to the advancement of high-energy and nuclear physics.

Know more

Different colours of fireworks (p.69)

The colours in fireworks come from the metal salts included. Together with fuels and other chemicals, metal salts are packed into small capsules. Once the fuel is ignited, it generates energy which excites the electrons in the atoms of the metal salt. When the electrons return to lower energy levels, photons are emitted. Since different elements have different sets of energy levels, different metal salts produce their own characteristic colours.



Metal salts in fireworks	Colour of fireworks
strontium carbonate	red
calcium chloride	orange
sodium nitrate	yellow
barium chloride	green
copper chloride	blue

Wave nature of electrons (p.80)

In 1927, *Clinton Davisson* and *Lester Germer* discovered that when electrons were fired at a nickel crystal, an electron diffraction pattern is observed. In the same year, *G. P. Thomson* also verified the wave nature of electrons by directing a beam of electrons at a thin metal foil. Davisson and Thomson were awarded the Nobel Prize in Physics in 1937.

Interestingly, G.P. Thomson's father, J. J. Thomson, was awarded the Nobel Prize in Physics due to the discovery of the particle nature of electrons.



Davisson (1881–1958) (left) and Germer (1896–1971).



G. P. Thomson (1892–1975).

Clinton Davisson 大衛遜　Lester Germer 革末　G. P. Thomson 喬治湯姆生